

Mean Field Control, Mean Field Games and Reinforcement Learning

Mathieu LAURIÈRE
NYU Shanghai

Slides:



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Mean-Field Games: Methodology And Practice*
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1. Introduction

- Background on Reinforcement Learning Methods
- From Multi-Agent Systems to Mean Field Control

2. RL for the General MFMDP Model

3. Policy Gradient for Linear-Quadratic MFC

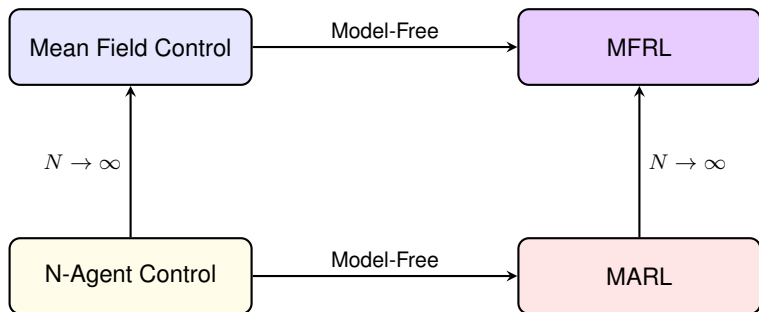
4. RL for Mean Field Games

5. Perspectives

Goal: Solve large-scale multi-agent problems.

Scalability:

- ▶ **Population Size:** via **Mean Field Approximation** ($N \rightarrow \infty$).
- ▶ **Environment Complexity:** via **(Deep) Reinforcement Learning** (Model-free).



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Markov Decision Process (MDP)

An MDP is defined by a tuple (S, A, P, c, γ) :

- ▶ **State Space** S : Set of possible situations.
- ▶ **Action Space** A : Set of possible decisions.
- ▶ **Transition Kernel** $P(x'|x, a)$: Probability of moving to x' given (x, a) .
Sometimes expressed with **transition function** F : $x' = F(x, a, \epsilon)$.
- ▶ **Cost Function** $c(x, a) \in \mathbb{R}$: Immediate cost after taking action a in state x .
- ▶ **Discount Factor** $\gamma \in [0, 1)$: Importance of future costs.

The Objective

Find a policy $\pi(a|x)$ that minimizes the **expected cumulative cost**:

$$J(\pi) = \mathbb{E}^{\pi} \left[\sum_{n=0}^{\infty} \gamma^n c(x_n, a_n) \right]$$

where $x_{n+1} \sim P(\cdot|x_n, a_n)$ and $a_n \sim \pi(\cdot|x_n)$.

Consider a single server in an environment.

- ▶ **States:** $x \in \{DI, DS, UI, US\}$.
 - ▶ **D/U:** Defended / Undefended.
 - ▶ **I/S:** Infected / Susceptible.
- ▶ **Action:** $a \in [0, 1]$ (level of protection).
- ▶ **Dynamics:** $x_{n+1} = F(x_n, a_n, \epsilon_n)$, where
 - ▶ a_n influences the probability of switching between levels of defense (D/U).
 - ▶ ϵ_n represents the risk of infection.
- ▶ **One-stage Cost:** $c(x, a) = k_D \mathbb{I}_{x \in \{DI, DS\}} + k_I \mathbb{I}_{x \in \{DI, UI\}}$.

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Remarks:

- ▶ For simplicity, the cost depends only on the state.
- ▶ Here, the transition depends only on the agent's own state and action. Later on we will include interactions with other agents.

- ▶ **State-Value Function:** $V^\pi(x) = \mathbb{E}^\pi \left[\sum_{k=0}^{\infty} \gamma^k c(x_{n+k}, a_{n+k}) \mid x_n = x \right]$.
- ▶ **Optimal Value:** $V^*(x) = \min_{\pi} V^\pi(x)$
- ▶ **Dynamic Programming:**

$$V^*(x) = \min_a \left\{ c(x, a) + \gamma \mathbb{E}[V^*(x')] \right\}.$$

- ▶ **Computation:** Value Iteration, ...

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Limitations:

- ▶ **✗ (L1):** From V^* only, we cannot recover an optimal policy.
- ▶ **✗ (L2):** Requires perfect knowledge of P and c .
- ▶ **✗ (L3):** Fails for large/continuous spaces due to *Curse of Dimensionality*.
- ▶ **✗ (L4):** Limited to one/a few agents.

State-Action Value Function

► State-Action Value (Q-function):

$$Q^\pi(x, a) = \mathbb{E}^\pi \left[\sum_{k=0}^{\infty} \gamma^k c(x_{n+k}, a_{n+k}) \mid x_n = x, a_n = a \right]$$

► Optimal Q-function: $Q^*(x, a) = \min_{\pi} Q^\pi(x, a)$.

Connection with V-function: $V^*(x) = \min_a Q^*(x, a)$

Bellman Equation

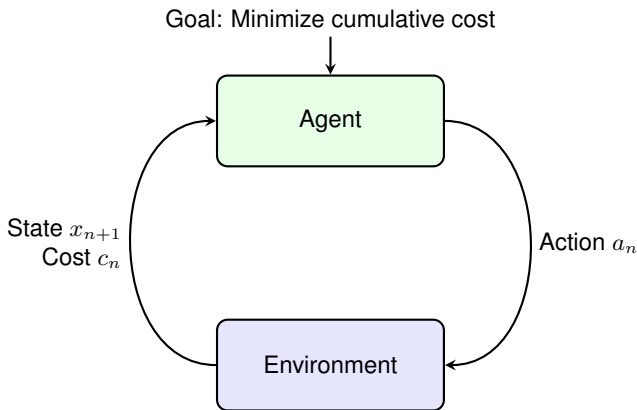
$$Q^*(x, a) = c(x, a) + \gamma \mathbb{E}_{x' \sim P(\cdot | x, a)} [\min_{a'} Q^*(x', a')]$$

Optimal Policy Recovery: If the optimal Q-function is known, an optimal policy can be obtained by greedy decisions:

$$\pi^*(x) \in \arg \min_{a \in A} Q^*(x, a) = \arg \min_{a \in A} \left\{ c(x, a) + \gamma \mathbb{E}_{x' \sim P(\cdot | x, a)} [V^*(x')] \right\}$$

✓ Answer to (L1)

The Reinforcement Learning Paradigm



- ▶ **Agent:** The learner/decision maker.
- ▶ **Environment:** Everything the agent interacts with.
- ▶ **State x_n :** Current situation of the environment.
- ▶ **Action a_n :** Choice made by the agent.
- ▶ **Cost $c_n = c(x_n, a_n)$:** One-step (running) cost.

- ▶ **Challenge:** Dynamics $P(x'|x, a)$ and costs $c(x, a)$ are often unknown
- ▶ **Goal:** Design *model-free* algorithms to learn Q^* or π^* .

Model-Free Learning: TD Learning

- ▶ **Challenge:** Dynamics $P(x'|x, a)$ and costs $c(x, a)$ are often unknown
- ▶ **Goal:** Design *model-free* algorithms to learn Q^* or π^* .
- ▶ **Temporal Difference (TD):** Update estimates based on individual transitions (x_n, a_n, c_n, x_{n+1}) .

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Q-Learning Update (Watkins, 1989)

The estimate $Q(x_n, a_n)$ is updated towards the **TD Target** $c_n + \gamma \min_{a'} Q(x_{n+1}, a')$:

$$Q(x_n, a_n) \leftarrow (1 - \eta_n)Q(x_n, a_n) + \eta_n \left(c_n + \gamma \min_{a'} Q(x_{n+1}, a') \right)$$

where $\eta_n \in (0, 1]$ is the learning rate.

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where $\eta_n \in (0, 1]$ is the learning rate.

- ▶ Converges to Q^* if all pairs (x, a) are visited infinitely often.
- ▶ Trade-off: **Exploration** (try new a) vs **Exploitation** (choose $\min_a Q$).
- ▶ Often: **Exploring starts** assumption.

✓ Answer to (L2)

- ▶ **Scalability Issue:** Tabular RL (storing Q in a table) fails when spaces S and/or A are very large or continuous (*Curse of Dimensionality*).
- ▶ We want to approximate Q as a function $S \times A \rightarrow \mathbb{R}$.
- ▶ **Solution:** Approximate $Q(x, a) \approx Q_\theta(x, a)$ using a neural network with parameters θ .

Motivations for Deep RL

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DQN (Deep Q-Network)

(Mnih et al., 2015)

Approximate Q^* using a neural net Q_θ by minimizing the loss:

$$\mathcal{L}(\theta) = \mathbb{E}_{(x,a,c,x') \sim \mathcal{D}} \left[\left(\underbrace{c + \gamma \min_{a'} Q_{\theta^-}(x', a')}_{\text{Target } y} - Q_\theta(x, a) \right)^2 \right]$$

- ▶ **Experience Replay:** Break correlations by sampling from a buffer $\mathcal{D}$.
- ▶ **Target Networks:** Target parameters θ^- are fixed between updates.

✓ Answer to (L3)

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- ▶ **Experience Replay:** Break correlations by sampling from a buffer $\mathcal{D}$.
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✓ Answer to (L3) but requires *finite* action space.

- **Idea:** Parameterize the policy $\pi_\theta(a|x)$ and optimize θ directly to minimize $J(\theta)$.

Policy Gradient Theorem

(Sutton et al., 1999)

The gradient $\nabla_\theta J(\theta)$ can be formulated as:

$$\nabla_\theta J(\theta) = \mathbb{E}_{\tau \sim \pi_\theta} \left[\sum_{n=0}^{\infty} \nabla_\theta \ln \pi_\theta(a_n | x_n) \cdot \left(\sum_{k=n}^{\infty} \gamma^{k-n} c_k \right) \right]$$

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- ▶ **Problem:** How to estimate the terms in the gradient?
- ▶ **REINFORCE:** Uses sample returns (trajectory-based).

✓ Answer to (L3) but suffers from high variance.

DDPG: Deep Deterministic Policy Gradient

Idea: Replace sum of future rewards by a learned Q -function.

Actor-Critic Architecture

- ▶ **Actor:** Represents the policy $\pi_{\phi}(x)$, mapping states to actions.
- ▶ **Critic:** Approx. the value function $Q_{\psi}(x, a)$, evaluating chosen actions.

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DDPG

(Lillicrap et al., 2016)

An off-policy algorithm designed for **continuous action spaces**:

- ▶ Uses a deterministic actor $a = \pi_\phi(x)$.
- ▶ Updates ϕ by following the gradient through the critic:

$$\nabla_\phi J \approx \mathbb{E} \left[\nabla_a Q_\psi(x, a)|_{a=\mu_\phi(x)} \nabla_\phi \pi_\phi(x) \right]$$

Note: For *stochastic* policies, see **SAC**, **PPO**, ...

✓ Answer to (L3): Scalable and handles continuous S and A .

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✗ How about (L4)? MFRL.

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Consider a system with $N \geq 1$ agents (finite population).

- ▶ **States:** $\underline{x}_n = (x_n^1, \dots, x_n^N) \in S^1 \times \dots \times S^N$.
- ▶ **Action Profile:** $\underline{a}_n = (a_n^1, \dots, a_n^N) \in A^1 \times \dots \times A^N$.
- ▶ **Policies:** Each agent i follows a policy $a_n^i = \pi_n^i(\underline{x}_n)$.
- ▶ **Policy Vector:** $\underline{\pi} = (\pi^1, \dots, \pi^N)$.
- ▶ **Objective:** Each agent i aims at minimizing:

$$J^i(\underline{\pi}) = \mathbb{E} \left[\sum_{n=0}^{\infty} \gamma^n f^i(\underline{x}_n, \underline{a}_n) \right]$$

Dynamics and Noise

- ▶ ϵ_n^i : **Idiosyncratic noise** (independent for each agent).
- ▶ ϵ_n^0 : **Common noise** (affects everyone simultaneously).
- ▶ Private state x^i evolves according to:

$$x_{n+1}^i = F^i(\underline{x}_n, \underline{a}_n, \epsilon_n^i, \epsilon_n^0)$$

Now consider N servers connected in a network.

- ▶ **State Vector:** $\underline{x}_n = (x_n^1, \dots, x_n^N)$.
- ▶ **Coupled Dynamics:** For server i , the evolution is:

$$x_{n+1}^i = F(x_n^i, a_n^i, \underline{x}_n, \epsilon_n^i)$$

where the **infection risk** for server i depends on the health status of **other** servers in the network (e.g., malware spread).

- ▶ **One-stage Cost:** $c^i(x^i, a^i) = k_D \mathbb{I}_{x^i \in \{DI, DS\}} + k_I \mathbb{I}_{x^i \in \{DI, UI\}}$.

To study the limit $N \rightarrow \infty$, we assume specific structures:

1. Symmetric Interaction (Anonymity)

The interaction depends only on the **empirical distribution** of states and actions:

► **Empirical State-Action Measure:** $\hat{\nu}_n^N = \frac{1}{N} \sum_{i=1}^N \delta_{(x_n^i, a_n^i)}.$

Then for all i ,

$$f^i(\underline{x}_n, \underline{a}_n) = f^i(x_n^i, a_n^i, \hat{\nu}_n^N), \quad F^i(\underline{x}_n, \underline{a}_n, \epsilon_n^i, \epsilon_n^0) = F^i(x_n^i, a_n^i, \hat{\nu}_n^N, \epsilon_n^i, \epsilon_n^0).$$

2. Homogeneity (Identical Agents)

All agents share the same spaces S, A , and the same cost and dynamics functions:

$$f^i = f, \quad F^i = F, \quad \text{for all } i \in \{1, \dots, N\}.$$

This structure ensures that agents are *interchangeable*.

The Optimization Landscape

Two major forms of multi-agent optimization:

1. Cooperation: Social Optimum

Agents cooperate to minimize the aggregate (average) cost:

$$\min_{\underline{\pi}} \tilde{J}(\underline{\pi}) = \min_{\underline{\pi}} \frac{1}{N} \sum_{i=1}^N J^i(\underline{\pi})$$

2. Competition: Nash Equilibrium

A profile $\underline{\pi}^*$ is a Nash equilibrium if no agent i can deviate alone to reduce their own cost:

$$J^i(\underline{\pi}^{*-i}, \pi') \geq J^i(\underline{\pi}^*) \quad \text{for any admissible } \pi'.$$

Focus: We will focus on the **cooperative** case.

By assuming symmetry and large N , the problem structure simplifies:

- ▶ **Symmetric Interaction:** The infection risk for a server depends only on the **proportion** of infected servers in the population:

$$\hat{\mu}_n^N = \frac{1}{N} \sum_{j=1}^N \delta_{x_n^j}$$

- ▶ **Cooperative Mean Field Setting:** The one-stage social cost is the integral over states and actions:

$$\frac{1}{N} \sum_{i=1}^N c(x_n^i, a_n^i) = k_D \hat{\mu}_n^N(\{DI, DS\}) + k_I \hat{\mu}_n^N(\{DI, UI\})$$

The total discounted cost to minimize is:

$$\mathbb{E} \left[\sum_{n=0}^{\infty} \gamma^n \left[k_D \hat{\mu}_n^N(\{DI, DS\}) + k_I \hat{\mu}_n^N(\{DI, UI\}) \right] \right]$$

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Note: We need $\mathbb{E}$ because $\hat{\mu}_n^N$ is stochastic. It becomes deterministic in the limit $N \rightarrow +\infty$ unless there is common noise.

Solving the N -agent cooperative problem directly is hard:

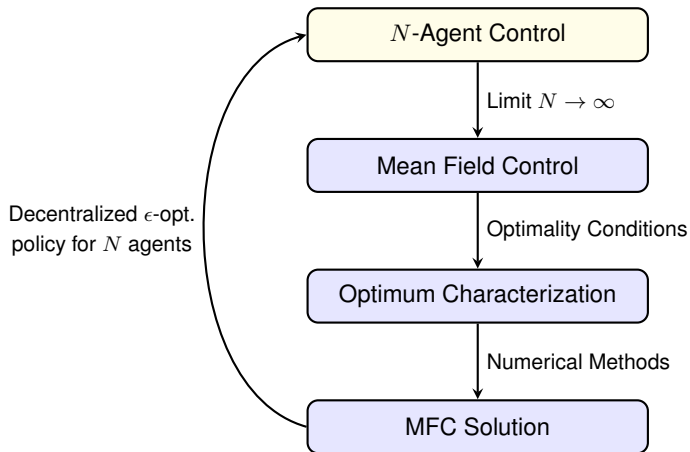
- ▶ **Curse of Dimensionality:** The state space $S^1 \times \dots \times S^N$ grows exponentially with N .
- ▶ **Policy Complexity:** Storing and computing N different policies $\underline{\pi} = (\pi^1, \dots, \pi^N)$ is prohibitive.

The Mean Field Idea

Under the **Symmetry** and **Homogeneity** assumptions:

- ▶ The problem simplifies as $N \rightarrow \infty$.
- ▶ We obtain a **Mean Field Control (MFC)** problem.
- ▶ We look for a *single* representative policy π instead of N .

The Mean Field Control Paradigm



Mean Field Control: Discrete-Time Model

In the limit $N \rightarrow \infty$, a conditional form of the **propagation of chaos** leads to conditional independence of individual dynamics.

Representative Agent Dynamics

The state of a representative agent x_n evolves according to:

$$x_{n+1} = F(x_n, a_n, \mathbb{P}_{(x_n, a_n)}^0, \varepsilon_{n+1}, \varepsilon_{n+1}^0), \quad n \geq 0,$$

where $\mathbb{P}_{(x_n, a_n)}^0$ stands for the conditional law of the state-action couple (x_n, a_n) given the common noise $(\varepsilon_n^0)_{n \geq 1}$.

MFC Optimization Problem

The objective in this limiting regime is:

$$\inf_{\pi} J(\pi), \quad J(\pi) = \mathbb{E} \left[\sum_{n=0}^{\infty} \gamma^n f(x_n, a_n, \mathbb{P}_{(x_n, a_n)}^0) \right].$$

This intuition is substantiated by rigorous mathematical consistency results.

Analogously to action randomization in standard MDPs, the central planner may use **common randomization**.

Common Randomization

- ▶ **Level-1:** The planner samples a policy π_n from a distribution using a common random variable (θ_n^0) . This π_n is shared by all agents.
- ▶ **Level-0:** Each agent samples an action a_n from $\pi_n(x_n)$ using individual randomization.
- ▶ The conditional distribution $\mathbb{P}^0$ is then conditioned on **both** the common noise (ε_n^0) and the common randomization (θ_n^0) .

Advantage: Allows exploration at the population-level.

Common Randomization and Sources of Stochasticity

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Common Randomization

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Summary: Four Types of Randomness

	Individual / Idiosyncratic	Common / Global
Noise	Agent-specific shocks ε^i	External environment ε^0
Randomization	Randomized local policy	Shared global policy sampling

The Lifted Mean-Field MDP (MFMDP)

For a policy π , the MFC value function J^π corresponds to an MDP on the space of measures. The **lifted MFMDP** is a six-tuple $(\bar{S}, \bar{A}, \bar{\Gamma}, P, \bar{f}, \gamma)$:

- ▶ **State Space:** $\bar{S} := \mathcal{P}(S)$. Generic element μ .
- ▶ **Action Space:** $\bar{A} := \mathcal{P}(S \times A)$. Generic element $\bar{a}$.
- ▶ **Graph:** $\bar{\Gamma} := \{(\mu, \bar{a}) \in \bar{S} \times \bar{A} \mid \bar{a} \in \bar{U}(\mu)\}$, where the **Control Constraint** is $\bar{U}(\mu) := \{\bar{a} \in \bar{A} \mid \text{pr}_1(\bar{a}) = \mu\}$.
- ▶ **Transition Kernel:** $\bar{P} : \bar{\Gamma} \rightarrow \mathcal{P}(\bar{S})$, a measurable function.
- ▶ **One-stage Cost:** $\bar{f} : \bar{\Gamma} \rightarrow \mathbb{R}$, bounded and measurable.
- ▶ **Discount factor:** $\gamma \in (0, 1)$.

The MFMDP is **consistent** with the MFC model $(S, A, E, E^0, F, f, \gamma)$ if:

► **Transition Kernel:**

$$\bar{P}(\mu, \bar{a})(d\mu') = (\nu^0 \circ \bar{F}(\mu, \bar{a}, \cdot)^{-1})(d\mu')$$

where the lifted system function $\bar{F}$ is defined by:

$$\bar{F}(\mu, \bar{a}, e^0) = (\bar{a} \otimes \nu) \circ F(\cdot, \cdot, \bar{a}, \cdot, e^0)^{-1}$$

► **Lifted Cost Function:**

$$\bar{f}(\mu, \bar{a}) = \int_{S \times A} f(x, a, \bar{a}) \bar{a}(dx, da)$$

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► **Lifted Cost Function:**

$$\bar{f}(\mu, \bar{a}) = \int_{S \times A} f(x, a, \bar{a}) \bar{a}(dx, da)$$

Optimal value function: $\bar{J}^*$; satisfies Dynamic Programming Principle.

At the mean field level:

- ▶ **Symmetric Interaction:** The infection risk for a server depends only on the **proportion** of infected servers in the population:

$$\mu_n = \mathbb{P}_{x_n}^0$$

- ▶ **Cooperative Mean Field Setting:** The one-stage social cost is the integral over states and actions:

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The total discounted cost to minimize is:

$$\mathbb{E} \left[\sum_{n=0}^{\infty} \gamma^n [k_D \mu_n(\{DI, DS\}) + k_I \mu_n(\{DI, UI\})] \right]$$

Note: We need $\mathbb{E}$ if there is common noise.

Two main frameworks when dealing with large populations:

- ▶ **Mean Field Control (MFC):** Social optimum
 - ▶ **Optimality:** Find a central policy π^* to minimize the average cost.
 - ▶ **Characteristic:** No incentive for joint deviations.
 - ▶ **Mathematical nature:** An **optimization** problem.

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- ▶ **Mean Field Games (MFG):** Nash equilibrium
 - ▶ **Optimality:** For a given mean field flow μ , each agent finds a best response $\hat{\pi}$.
 - ▶ **Consistency:** The resulting mean field flow from all agents using $\hat{\pi}$ must be exactly μ .
 - ▶ **Characteristic:** No incentive for unilateral deviations.
 - ▶ **Mathematical nature:** A **fixed point** problem over the mean field μ .

Two main frameworks when dealing with large populations:

- ▶ **Mean Field Control (MFC):** Social optimum
 - ▶ **Optimality:** Find a central policy π^* to minimize the average cost.
 - ▶ **Characteristic:** No incentive for joint deviations.
 - ▶ **Mathematical nature:** An **optimization** problem.

- ▶ **Mean Field Games (MFG):** Nash equilibrium
 - ▶ **Optimality:** For a given mean field flow μ , each agent finds a best response $\hat{\pi}$.
 - ▶ **Consistency:** The resulting mean field flow from all agents using $\hat{\pi}$ must be exactly μ .
 - ▶ **Characteristic:** No incentive for unilateral deviations.
 - ▶ **Mathematical nature:** A **fixed point** problem over the mean field μ .

Focus: The tutorial is mainly focused on MF Control (cooperative), but we will present RL algorithms for MF Games (non-cooperative) later on.

Individual Cost in MFG: In contrast to MFC where $J(\pi)$ is the average population cost, the cost for a representative agent in an MFG scenario depends on their own policy π and the population's behavior.

- ▶ **As a function of a mean field flow:** $J(\pi; \mu)$, where $\mu = (\mu_n)_{n \geq 0}$ is the population's state distribution sequence over time.

MFG Individual Cost Function

Given a population flow μ , the expected cumulative cost for a representative agent using policy π is:

$$J(\pi; \mu) = \mathbb{E} \left[\sum_{n=0}^{\infty} \gamma^n f(x_n, a_n, \mu_n) \right]$$

where individual dynamics are determined by $a_n \sim \pi(\cdot | x_n)$ and the environment F .

Mean Field Games (MFG): Population Policy

Alternatively, we can define the cost assuming the population uses a shared policy π' .

- ▶ Individual cost rewritten as $J(\pi; \pi')$, where π is the agent's policy and π' is the population's policy.
- ▶ This is evaluated by setting $\mu = \mu^{\pi'}$, the distribution flow generated by π' .

Population Distribution Evolution

The sequence $\mu^{\pi'} = (\mu_n^{\pi'})_{n \geq 0}$ follows a Markov chain-type evolution: Assuming no common noise,

$$\mu_{n+1}^{\pi'}(x') = \sum_{x \in S} \sum_{a \in A} P(x'|x, a) \pi'(a|x) \mu_n^{\pi'}(x)$$

starting from an initial distribution $\mu_0^{\pi'} = \mu_0$. Thus, we define:

$$J(\pi; \pi') := J(\pi; \mu^{\pi'})$$

MFG Individual Cost Reformulation

With the above notation, we define:

$$J(\pi; \pi') := J(\pi; \mu^{\pi'})$$

Mean Field Nash Equilibrium (MFNE) Definition

For a fixed population behavior μ , the agent aims to find a **best response** policy $\hat{\pi}$ by solving a standard Markov Decision Process: $\hat{\pi} \in \arg \min_{\pi} J(\pi; \mu)$.

A Mean Field Nash Equilibrium (MFNE) characterizes a state where no agent has an incentive to unilaterally deviate. It can be formulated in two equivalent ways:

Definition 1 (Policy-based)

A policy $\hat{\pi}$ is a Mean Field Nash Equilibrium if it is a best response against itself:

$$\hat{\pi} \in \arg \min_{\pi} J(\pi; \hat{\pi})$$

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Definition 2 (Policy-Distribution pairs)

A pair $(\hat{\pi}, \hat{\mu})$ is a Mean Field Nash Equilibrium if it satisfies exactly two conditions:

1. **Optimality (Best Response):** $\hat{\pi}$ is optimal given the fixed population $\hat{\mu}$:

$$\hat{\pi} \in \arg \min_{\pi} J(\pi; \hat{\mu})$$

2. **Consistency (Mean Field):** The distribution flow $\hat{\mu}$ is exactly the one generated when the whole population uses policy $\hat{\pi}$:

$$\hat{\mu}_n = \mathbb{P}_{x_n^{\hat{\pi}}} \quad \text{for all } n \geq 0$$

- ▶ **MFG foundations:** Lasry–Lions [[LL07a](#)]; Nash certainty equivalence [[HMC06](#)]; monographs and lecture notes [[BFY⁺13](#), [CD18a](#), [CD18b](#), [ACD⁺20](#)].
- ▶ **Mean field control and MFMDP:** population MDP viewpoint [[GG11](#), [GGB12](#)]; probabilistic MFC and McKean–Vlasov control [[CLT23a](#), [CD18a](#), [MP22](#)]; dynamic programming for MFMDPs [[CLT23b](#), [MP23](#)].
- ▶ **Single-agent and multi-agent RL background:** stochastic control and MDPs [[BS96](#), [Put14](#)]; MARL surveys [[CTE⁺22a](#), [ZYB21](#)].
- ▶ **Numerics and learning surveys:** numerical methods for MFG/MFC [[Lau21](#)]; machine learning for stochastic control/games [[HL22](#)]; learning in MFGs [[LPP⁺](#)].

- ▶ **Crowds, flocking, mobility:** crowd motion and routing [ALL19, PPL+20, CLP+22]; flocking [NCM11, PLP+21]; urban mobility and traffic [DXS25, MCD+24].
- ▶ **Economics and finance:** early application surveys [GLL11]; systemic risk and finance/economics [CS15, Car21, BVTR23]; social networks [NM18].
- ▶ **Energy, security, communications:** energy systems [ATM20, EHMP19, GB18]; security and wireless communications [MVL12, SBS+15, YLS+17, KM18].
- ▶ **Engineering and autonomous systems:** autonomous vehicles and traffic control [HDDC17, HDDC19, TNPJ20]; safe mean-field MARL [JPJ+24].

Outline

1. Introduction

2. RL for the General MFMDP Model

- Setup
- Mean Field Q-Learning
- Deep Reinforcement Learning
- Numerical Illustrations

3. Policy Gradient for Linear-Quadratic MFC

4. RL for Mean Field Games

5. Perspectives

This part is mostly based on:

Carmona, Lauriere, Tan: AAP 2023

and lecture notes

Carmona & Lauriere: Mean Field Reinforcement Learning
(arXiv:2607.01525)

Outline

1. Introduction

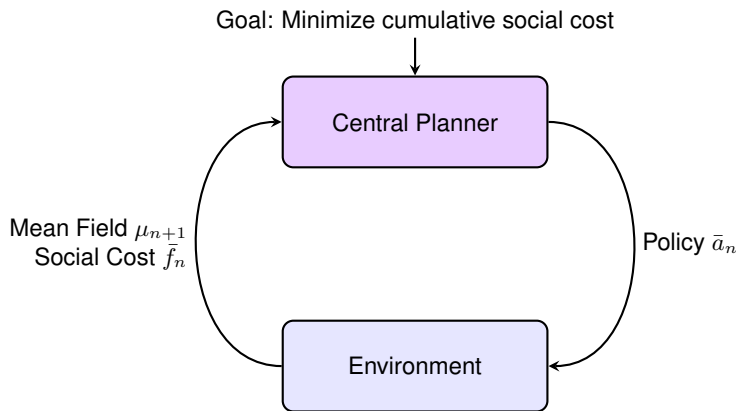
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- ▶ **Central Planner:** Samples a policy shared by the population.
- ▶ **Environment:** Global dynamics of the population.
- ▶ **Mean Field μ_n :** State of the population (distribution).
- ▶ **Policy $\bar{a}_n$:** Corresponds to a Level-1 action in the MFMDP.
- ▶ **Social Cost $\bar{f}_n = \bar{f}(\mu_n, \bar{a}_n)$:** Aggregate population cost.

- ▶ Assume individual spaces S and A are finite.
- ▶ $\mathcal{P}(S)$ and $\mathcal{P}(A)$ are simplices.
- ▶ Marginal distribution constraints:

$$\sum_a \bar{a}(x, a) = \mu(x).$$

- ▶ Central Planner Actions:

$$\tilde{A} := \{\tilde{a} : S \rightarrow \mathcal{P}(A) \mid \tilde{a} \text{ Borel measurable}\}$$

- ▶ Joint distribution: $\mu \hat{\otimes} \tilde{a}$.
- ▶ Q-Function rewritten:

$$\tilde{Q}^*(\mu, \tilde{a}) := \bar{Q}^*(\mu, \mu \hat{\otimes} \tilde{a}).$$

Example: Cybersecurity Model Revisited

- ▶ **Scenario:** A central planner protects a large network of computers.
- ▶ **Individual States:** Each computer is in one of 4 states ($|S| = 4$):
 - ▶ **D**efended or **U**ndefended
 - ▶ **I**nfected or **S**usceptible
- ▶ **Dynamics:**
 - ▶ Computers recover from infection at fixed rates.
 - ▶ New infections occur via *external* hackers or *internal* spread.
 - ▶ **Mean Field Interaction:** Internal spread rates (e.g., $\beta\mu(\{DI\})$) depend on the percentage of infected neighbors.
- ▶ **Goal:** Minimize protection costs + infection penalties across the population.

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Mathematically:

- ▶ **State Set:** $S = \{DI, DS, UI, US\}$.
- ▶ **Transitions:** The intensity of transitioning from $DS \rightarrow DI$ is:

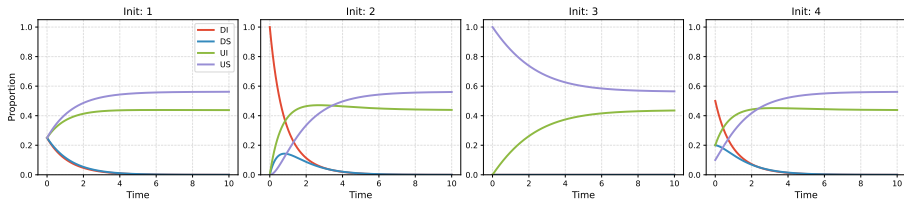
$$P_{DS \rightarrow DI}^{\mu, a} = v_H q_{inf}^D + \beta_{DD}\mu(\{DI\}) + \beta_{UD}\mu(\{UI\})$$

- ▶ **Total Cost:** The central planner pays for defense and infection:

$$\bar{f}(\mu, \bar{a}) = k_D\mu(\{DI, DS\}) + k_I\mu(\{DI, UI\})$$

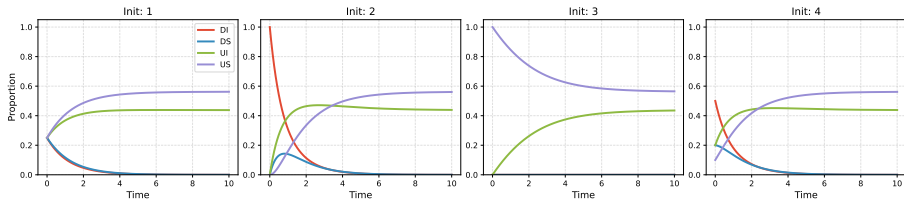
Continuous-time benchmark:

- ▶ solution through forward-backward ODEs
- ▶ for each *given* initial distribution



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- ▶ solution through forward-backward ODEs
- ▶ for each *given* initial distribution



Goal: learn the optimal action

- ▶ for every individual state
- ▶ for *every* mean field distribution

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Simplex Discretization and Tabular MFQ-Learning

- ▶ $\mathcal{P}(S)$ and $\mathcal{P}(A)$ are infinite continuous simplices.
- ▶ We want to start with a **simple tabular RL method** for finite spaces.
- ▶ **Idea:** Discretize $\mathcal{P}(S)$ and $\mathcal{P}(A)$ onto finite grids: $\check{\mathcal{S}} \subset \mathcal{P}(S)$ and $\check{\mathcal{A}} \subset \mathcal{P}(A)$.

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- ▶ **Resolution Parameters:**
 - ▶ $\varepsilon_{\check{\mathcal{S}}}$: State grid mesh size (e.g., $1/N_m$ where N_m is the mesh divisions).
 - ▶ $\varepsilon_{\check{\mathcal{A}}}$: Action grid mesh size.
- ▶ **Cost Function:** $\tilde{f}(\mu, \tilde{a}) = \bar{f}(\mu, \mu \hat{\otimes} \tilde{a})$ be the planner's cost given $\tilde{a} : S \rightarrow \mathcal{P}(A)$.
- ▶ **Projection:** At each step, project onto the grid:

$$\check{\mu}_{n+1} = \text{proj}_{\check{\mathcal{S}}} \circ \bar{F}(\check{\mu}_n, \check{\mu}_n \hat{\otimes} \check{a}, \varepsilon_{n+1}^0)$$

(Alternatively: project only when updating the Q-table)

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Cybersecurity example:

- ▶ **State Simplex:** $\mathcal{S} = \{(\mu^{(1)}, \dots, \mu^{(4)}) \in [0, 1]^4 \mid \sum_i \mu^{(i)} = 1\}$.
- ▶ **Discretized Grid:** $\check{\mathcal{S}} = \{(\mu^{(i)}) \in \{0, 1/N_m, \dots, 1\}^4 \mid \sum_i \mu^{(i)} = 1\}$.

The Curse of Dimensionality: $|\check{\mathcal{S}}| = \binom{N_m + |S| - 1}{|S| - 1}$, poly. in N_m but *expo.* in $|S|$:

$N_m \setminus S $	2	3	4	5
10	11	66	286	1,001
20	21	231	1,771	10,626
50	51	1,326	23,426	316,251

Algorithm: Tabular MFQ-Learning with Discretization

Input: Grids $\check{\mathcal{S}}, \check{A}$, episodes N_{total} , length N_{epi} , learning rate η .

Output: Trained Q-table $\check{Q}$.

```
1 Initialize table  $\check{Q}_0$  on  $\check{\mathcal{S}} \times \check{A}$ 
2 for  $m = 1, \dots, N_{total}$  do
3     Sample initial distribution  $\mu_0 \in \mathcal{S}$  and set  $\check{\mu}_0 = \text{proj}_{\check{\mathcal{S}}}(\mu_0)$ 
4     for  $n = 0, \dots, N_{epi} - 1$  do
5         Select action  $\check{a}_n$  from  $\check{\mu}_n$  (e.g., using  $\epsilon$ -greedy)
6         Execute  $\check{a}_n$ , observe cost  $\tilde{f}(\check{\mu}_n, \check{a}_n)$  and new state:
7          $\check{\mu}_{n+1} = \text{proj}_{\check{\mathcal{S}}} \circ \bar{F}(\check{\mu}_n, \check{\mu}_n \otimes \check{a}_n, \varepsilon_{n+1}^0)$ 
8         Update:
9         
$$\check{Q}(\check{\mu}_n, \check{a}_n) \leftarrow (1 - \eta_k) \check{Q}(\check{\mu}_n, \check{a}_n) + \eta_k \left( \tilde{f}(\check{\mu}_n, \check{a}_n) + \gamma \min_{a'} \check{Q}(\check{\mu}_{n+1}, a') \right)$$

10 return  $\check{Q}$ 
```

Theorem (MFQ-Learning Convergence (Informal))

The learned table $\check{Q}$ converges to the true infinite-population Q -function, up to:

- ▶ *tabular Q -learning error*
- ▶ *discretization error.*

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Extra: Learned (greedy) policy converges.

Two scenarios:

▶ **Mixed Controls (Level-0 Randomization):**

- ▶ Representative agent uses randomized actions: $\tilde{a} : S \rightarrow \mathcal{P}(A)$.
- ▶ $\tilde{A} \approx \mathcal{P}(A)^{|S|}$ requires discretizing both state and action simplices.
- ▶ Error bound ε' depends on both grid resolutions ε_S and ε_A .

▶ **Pure Controls (Deterministic Level-0):**

- ▶ In some MFC problems, optimal policies are non-randomized at level-0.
- ▶ Representative agent uses deterministic actions: $\tilde{a} : S \rightarrow A$.
- ▶ Action space $\tilde{A} \approx A^S$ is finite; no discretization of $\mathcal{P}(A)$ needed.
- ▶ Computationally cheaper: updating a table of size $|\check{S}| \times |A|^{|S|}$.

Regularity and Exploration Assumptions

- ▶ **Lipschitz Data:** Cost $\bar{f}$ and dynamics $\bar{F}$ are Lipschitz in $(\mu, \tilde{a})$.
- ▶ **Value Function:** Optimal $\mu \mapsto \bar{J}^*(\mu)$ is Lipschitz continuous.
- ▶ **Covering Time:** $(\check{\mu}_n, \check{a}_n)$ covers the grid $\check{\mathcal{S}} \times \check{A}$.
- ▶ **Learning Rates:** $\eta_n = 1/(1 + C(n, \check{\mu}, \check{a}))$, with $C(n, \check{\mu}, \check{a}) = \text{nb of visits}$.

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Theorem (Q-Table Convergence (Informal))

Let $\delta \in (0, 1)$ and $\varepsilon > 0$. If the total number of steps is of order

$$N_{\text{total}} \times N_{\text{epi}} = \Omega \left(\frac{\ln(1/\delta)^5}{\varepsilon^2} \right),$$

then with probability $1 - \delta$, for all $(\mu, \tilde{a}) \in \mathfrak{S} \times \tilde{\mathcal{A}}$,

$$\left| \check{Q}_{N_{\text{total}}} \left(\text{Proj}_{\check{\mathfrak{S}}}(\mu), \text{Proj}_{\check{\mathcal{A}}}(\tilde{a}) \right) - \bar{Q}^*(\mu, \mu \hat{\otimes} \tilde{a}) \right| \leq \varepsilon',$$

where $\varepsilon' = \mathcal{O} \left(\varepsilon + \left(\frac{1}{1-\gamma} + 1 \right) \varepsilon_{\mathfrak{S}} + \frac{1}{1-\gamma} \varepsilon_{\mathfrak{A}} \right)$.

► **Covering time:**

- $T_{cov} = |\check{\mathcal{G}} \times \check{\mathcal{A}}|$ if full sweep
- In practice: follow trajectory
- Enough to assume that the covering is smaller than L with probability $1/2$.

► **Regularity:** Extension to the case where $\bar{J}^*$ is only p -Hölder continuous.

Detailed Assumptions

Assumption: Regularity of the data

$\tilde{f}$ is bounded and Lipschitz continuous with respect to $(\mu, \tilde{a})$ with constant $L_{\tilde{f}}$, namely for every $(\mu, \tilde{a}), (\mu', \tilde{a}') \in \mathfrak{S} \times \tilde{\mathcal{A}}$, we have

$$|\tilde{f}(\mu, \tilde{a}) - \tilde{f}(\mu', \tilde{a}')| \leq L_{\tilde{f}} (\|\mu - \mu'\|_{d_{\mathfrak{S}}} + d_{\tilde{\mathcal{A}}}(\tilde{a}, \tilde{a}')) \quad \text{and} \quad \tilde{f}(\mu, \tilde{a}) \leq L_{\tilde{f}}.$$

Also, $\bar{F}$ is Lipschitz continuous with respect to μ and $\tilde{a}$ with constant $L_{\bar{F}}$ in expectation over the randomness of the common noise, namely for every $(\mu, \tilde{a}), (\mu', \tilde{a}') \in \mathfrak{S} \times \tilde{\mathcal{A}}$:

$$\mathbb{E}_{\varepsilon^0} [\|\bar{F}(\mu, \mu \hat{\otimes} \tilde{a}, \varepsilon^0) - \bar{F}(\mu', \mu' \hat{\otimes} \tilde{a}', \varepsilon^0)\|_{d_{\mathfrak{S}}}] \leq L_{\bar{F}} (\|\mu - \mu'\|_{d_{\mathfrak{S}}} + d_{\tilde{\mathcal{A}}}(\tilde{a}, \tilde{a}')).$$

Assumption: Regularity of the value function

$\bar{J}^*$ is Lipschitz continuous w.r.t. μ with constant $L_{\bar{J}^*}$.

Assumption: Covering time

There exists a finite T_{cov} such that for every n_0 , $\check{\mathfrak{S}} \times \check{\mathcal{A}} \subseteq (\check{\mu}_n, \check{a}_n)_{n=n_0, \dots, n_0+T_{cov}}$.

Assumption: Learning rate

There exists $\kappa \in (1/2, 1)$ such that for every $(\check{\mu}, \check{a}) \in \check{\mathfrak{S}} \times \check{\mathcal{A}}$, $\eta_n := \eta_n(\check{\mu}, \check{a}) = 1/(1 + C(n, \check{\mu}, \check{a}))^\kappa$ for each $n \geq 0$, where $C(n, \check{\mu}, \check{a})$ is the number of times up to n that the pair $(\check{\mu}, \check{a})$ has been visited in the MFQ-Learning Algorithm.

- ▶ $\check{J}_{bound}$: upper bound on the optimal value of the projected MFC problem.

- ▶ $\text{Proj}_{\check{\mathfrak{S}}} : \mathfrak{S} \rightarrow \check{\mathfrak{S}}$ and $\text{Proj}_{\check{\mathcal{A}}} : \tilde{\mathcal{A}} \rightarrow \check{\mathcal{A}}$: projection operators such that

$$(\text{Proj}_{\check{\mathfrak{S}}}(\mu), \text{Proj}_{\check{\mathcal{A}}}(\tilde{a})) := (\check{\mu}, \check{a})$$

for every $(\mu, \tilde{a}) \in \mathfrak{S} \times \tilde{\mathcal{A}}$ where $(\check{\mu}, \check{a})$ is the closest point (or one of the closest points, in case of equality) in $\check{\mathfrak{S}} \times \check{\mathcal{A}}$ with respect to $d_{\mathfrak{S}}$ and $d_{\tilde{\mathcal{A}}}$.

- ▶ Based on simplex discretizations:

$$\|\mu - \check{\mu}\|_{d_{\mathfrak{S}}} \leq \varepsilon_{\mathfrak{S}}, \quad d_{\tilde{\mathcal{A}}}(\tilde{a}, \check{a}) \leq \varepsilon_{\mathfrak{A}}.$$

- ▶ $\beta = (1 - \gamma)/2$ the horizon of the MFMDP

- ▶ For $\delta \in (0, 1)$, we let $T_{cov}(\delta) = \lceil T_{cov} \log_2(1/(2\delta)) \rceil$.

Theorem (Q-Table Convergence)

Let $\delta \in (0, 1)$ and $\varepsilon > 0$. If the total number of steps is of order

$$\Omega \left(\left(\frac{(T_{cov}(\delta))^{1+3\kappa} \check{J}_{bound}^2 \ln(|\check{\mathfrak{S}}| |\check{\mathfrak{A}}|^{|S|} \check{J}_{bound} / (2\delta\beta\varepsilon))}{\beta^2 \varepsilon^2} \right)^{\frac{1}{\kappa}} + \left(\frac{(T_{cov}(\delta))}{\beta} \ln \left(\frac{\check{J}_{bound}}{\varepsilon} \right) \right)^{\frac{1}{1-\kappa}} \right),$$

then with probability $1 - \delta$, for all $(\mu, \tilde{a}) \in \mathfrak{S} \times \tilde{\mathcal{A}}$,

$$\left| \check{Q}_{N_{total}} \left(\text{Proj}_{\check{\mathfrak{S}}}(\mu), \text{Proj}_{\tilde{\mathcal{A}}}(\tilde{a}) \right) - \bar{Q}^*(\mu, \mu \hat{\otimes} \tilde{a}) \right| \leq \varepsilon',$$

where $\varepsilon' = \varepsilon + \left(\frac{\gamma}{1-\gamma} L_{\bar{J}^*} + L_{\bar{f}} + \gamma L_{\bar{J}^*} L_{\bar{F}} \right) \varepsilon_{\mathfrak{S}} + \frac{1}{1-\gamma} \left(L_{\bar{f}} + \gamma L_{\bar{J}^*} L_{\bar{F}} \right) \varepsilon_{\mathfrak{A}}$.

We decompose the error using the triangle inequality into three manageable terms:

$$\begin{aligned} \left| \check{Q}(\hat{\mu}, \hat{a}) - \tilde{Q}^*(\mu, \tilde{a}) \right| &\leq \underbrace{\left| \check{Q}(\hat{\mu}, \hat{a}) - \check{Q}^*(\hat{\mu}, \hat{a}) \right|}_{\text{Step 1: Q-Learning Convergence}} \\ &\quad + \underbrace{\left| \check{Q}^*(\hat{\mu}, \hat{a}) - \tilde{Q}^*(\hat{\mu}, \hat{a}) \right|}_{\text{Step 2: Discretization Stability}} \\ &\quad + \underbrace{\left| \tilde{Q}^*(\hat{\mu}, \hat{a}) - \tilde{Q}^*(\mu, \tilde{a}) \right|}_{\text{Step 3: Continuous Regularity}} \end{aligned}$$

where $(\hat{\mu}, \hat{a}) = (\text{proj}_{\mathcal{S}}(\mu), \text{proj}_{\tilde{A}}(\tilde{a}))$.

1. **Step 1:** Standard asynchronous RL results (e.g. Even-Dar & Mansour):

$$\check{Q} \rightarrow \check{Q}^* \text{ as nb of steps} \rightarrow +\infty.$$

2. **Step 2:** Stability under projection.
3. **Step 3:** Continuous regularity: uses Lipschitz properties of $\tilde{f}$ and $\tilde{J}^*$.

If agents use pure controls: no need to discretize $\mathcal{P}(\mathcal{A})$.

Theorem (Q-Table Convergence with Pure Level-0 Controls)

If the total steps is sufficiently large, the error bound simplifies:

$$\sup_{\mu, \tilde{a} \in A^S} \left| \check{Q}(\text{proj}_{\check{\mathcal{S}}}(\mu), \tilde{a}) - \tilde{Q}^*(\mu, \delta_{\tilde{a}}) \right| \leq \varepsilon',$$

where:

$$\varepsilon' = \mathcal{O} \left(\varepsilon + \left(\frac{1}{1-\gamma} + 1 \right) \varepsilon_{\mathcal{S}} \right)$$

Policy Convergence and Action Gap

Recovering a control from a value function:

- ▶ Greedy choice: take the argmin of the Q-function
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Assumption: Action Gap

There exists $K_A > 0$ s.t. the optimal action is better than the second-best by K_A :

$$\tilde{Q}^*(\check{\mu}, \tilde{a}^*) - \check{Q}^*(\check{\mu}, \tilde{a}) \geq K_A, \quad \forall \tilde{a} \neq \tilde{a}^*$$

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Corollary: Policy Convergence

Under the above assumption, then:

$$\left\| \text{softmin}_\tau(\check{Q}(\check{\mu}, \cdot)) - \text{argmine}(\tilde{Q}^*(\check{\mu}, \cdot)) \right\|_2 \leq \tau \varepsilon' \sqrt{|\tilde{A}|} + 2e^{-\tau K_A} |\tilde{A}|$$

Notation: $\text{softmin}_\tau : \mathbb{R}^m \rightarrow \mathbb{R}^m$ by $\text{softmin}_\tau(x) = (e^{-\tau x_1}, \dots, e^{-\tau x_m}) / \sum_j e^{-\tau x_j}$, and argmine is the uniform distribution over the set of minimizers.

Policy Convergence and Action Gap

Recovering a control from a value function:

- ▶ Greedy choice: take the argmin of the Q-function
- ▶ *Does it converge?* Not obvious!

We make use of an extra assumption.

Assumption: Action Gap

There exists $K_A > 0$ s.t. the optimal action is better than the second-best by K_A :

$$\tilde{Q}^*(\check{\mu}, \tilde{a}^*) - \check{Q}^*(\check{\mu}, \tilde{a}) \geq K_A, \quad \forall \tilde{a} \neq \tilde{a}^*$$

Corollary: Policy Convergence

Under the above assumption, then:

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Tradeoff with τ (inverse temperature):

- ▶ $\tau \uparrow$ implies more concentrated but upper bound $\uparrow$

- ▶ Here we assume perfect simulation of the mean field.
- ▶ What if the mean field is *imperfectly* computed?
- ▶ For example: obtained by simulating a population of N agents.
⇒ Extra approximation error (**propagation of chaos**)
- ▶ See later: analysis of convergence of Policy Gradient in LQ MFC.

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5. Perspectives

Issue with Discretization: Curse of Dimensionality

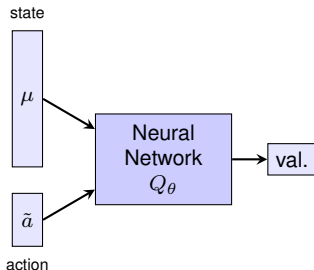
⇒ Discretizing the population state $\mathcal{P}(S)$ is intractable for large $|S|$.

Issue with Discretization: Curse of Dimensionality

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Neural Network as function approximator:

- ▶ Replace the Q-table with a network $Q_\theta(\mu, \tilde{a})$.
- ▶ Example: **Deep Q-Networks (DQN)**:
 - ▶ *Input*: Current distribution μ .
 - ▶ *Output*: Level-1 action $\tilde{a}$.

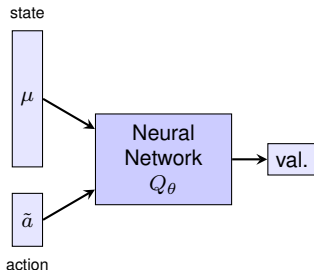


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Challenge:

- ▶ DQN requires finding $\min_{\tilde{a}} Q_\theta(\mu, \tilde{a})$, which is hard in continuous spaces.
- ▶ In MFC, $\tilde{A}$ is often continuous (e.g., the simplex $\mathcal{P}(A)$).

► **Deterministic Policy (Level-1):**

- For MFMDP, we have shown that there is an optimal policy which is pure or “deterministic”
- “Deterministic” refers to the *central planner’s choice* of the level-1 action $\tilde{a}$.
- The individual agents (level-0) can still follow stochastic strategies if $\tilde{a} \in \mathcal{P}(A)^S$.

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► **Deep Deterministic Policy Gradient (DDPG):**

- Designed to handle continuous action spaces directly.
- Relies on an **Actor** to output the action and a **Critic** to evaluate it.

To train the model, we use the following architectures:

- ▶ **Critic Network** $Q_\psi(\mu, \tilde{a})$: Approximates the optimal Q-value (Q^*).
- ▶ **Actor Network** $\pi_\phi(\mu)$: Approximates the optimal level-1 policy (π^*).
- ▶ **Experience Replay Buffer** $\mathcal{R}$: Stores past transitions $(\mu_n, \tilde{a}_n, \tilde{f}_n, \mu_{n+1})$.
- ▶ **Target Networks** $Q_{\tilde{\psi}}, \pi_{\tilde{\phi}}$: Slowly updated to stabilize training.

Algorithm: DDPG for MFMDPs

Input: Actor π_ϕ , Critic Q_ψ , Replay Buffer $\mathcal{D}$, episodes N_{total} , length N_{epi} .

Output: Optimized network parameters ϕ^*, ψ^* .

```
1 Initialize weights  $\phi, \psi$ ; sets targets  $\tilde{\phi} \leftarrow \phi, \tilde{\psi} \leftarrow \psi$ ; initialize  $\mathcal{R}$ 
2 for  $m = 1, \dots, N_{total}$  do
3     Sample  $\mu_0$ 
4     for  $n = 0, \dots, N_{epi} - 1$  do
5         Select action  $\tilde{a}_n = \pi_\phi(\mu_n) + \epsilon_n^a, \epsilon_n^a \sim \mathcal{N}$ 
6         Execute  $\tilde{a}_n$ , observe cost  $\tilde{f}_n$  and new state  $\mu_{n+1}$ 
7         Store transition  $(\mu_n, \tilde{a}_n, \tilde{f}_n, \mu_{n+1})$  in  $\mathcal{R}$ 
8         Sample minibatch from  $\mathcal{R}$ 
9         Set target:  $y_i = \tilde{f}_i + \gamma Q_{\tilde{\psi}}(\mu'_{i+1}, \pi_{\tilde{\phi}}(\mu'_{i+1}))$ 
10        Update Critic by minimizing loss:  $\mathcal{L}(\psi) = \frac{1}{B} \sum_i (y_i - Q_\psi(\mu_i, \tilde{a}_i))^2$ 
11        Update Actor by policy gradient:
12            
$$\nabla_\phi J(\phi) \approx \frac{1}{B} \sum_i \nabla_a Q_\psi(\mu_i, a)|_{a=\pi_\phi(\mu_i)} \nabla_\phi \pi_\phi(\mu_i)$$

13        Soft update targets:  $\tilde{\phi}, \tilde{\psi} \leftarrow \tau(\phi, \psi) + (1 - \tau)(\tilde{\phi}, \tilde{\psi})$ 
```

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We illustrate the RL algorithms on two testbeds:

1. Cybersecurity Model:

- ▶ $|S| = 4$ and $|A| = 2$.
- ▶ **Goal:** Protect system against external hackers and internal spread.
- ▶ **Benchmark** problem with reference solution.
- ▶ Pure controls at the level-0.
- ▶ **Solvers:** Tabular Q-Learning (Sync/Async) and MF-DDPG.

2. Discrete Distribution Planning:

- ▶ Larger state space ($|S| = 11$) and action space ($|A| = 3$).
- ▶ **Goal:** Move population mass to match a target configuration.
- ▶ Mixed controls at the level-0.
- ▶ **Solver:** MF-DDPG only (scales better).

- ▶ **Individual States:** $S = \{DI, DS, UI, US\}$ (Defended/Infected, Susceptible).
- ▶ **Planner's Goal:** Protect system against external hackers and internal spread.
- ▶ **Transition Matrix** $P^{\mu,a}$:

$$\begin{pmatrix} \dots & P_{DS \rightarrow DI}^{\mu,a} & \lambda a & 0 \\ q_{rec}^D & \dots & 0 & \lambda a \\ \lambda a & 0 & \dots & P_{US \rightarrow UI}^{\mu,a} \\ 0 & \lambda a & q_{rec}^U & \dots \end{pmatrix}$$

where infection rates $P_{DS \rightarrow DI}$ depend on attack intensity v_H and infected neighbors $\mu(\{DI\}), \mu(\{UI\})$.

- ▶ **Time discretization step:** Δt (to approximate continuous-time Markov chains).
- ▶ **Social Cost** $\bar{f}(\mu, \bar{a}) = k_D \mu(\{DI, DS\}) + k_I \mu(\{DI, UI\})$.

Ex. 1: Cybersecurity – Tabular Q-Learning Solvers

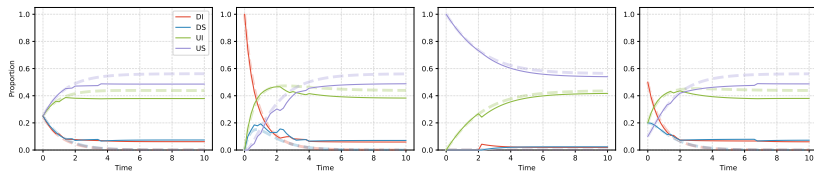
We evaluate three training schemes for the table $\check{Q}(\check{\mu}, \check{a})$:

1. **SyncFull**: Updated for all $(\check{\mu}, \check{a})$.
2. **SyncGreedy**: Synchronous update for every state $\check{\mu}$, but using only a single sample for the ϵ -greedy action.
3. **AsyncTraj**: Standard trajectory-based update (Standard Q-learning).

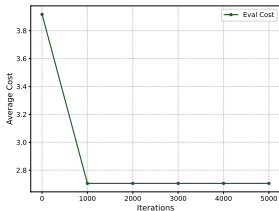
Param	Value	Param	Value
N_m	50	v_H	0.6
Δt	0.2	k_D, k_I	0.3, 0.5
T	10.0	q_{rec}^D, q_{rec}^U	0.5, 0.4

Cybersecurity parameters.

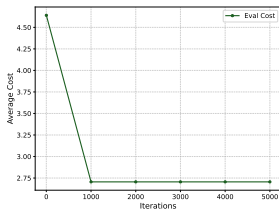
Ex. 1: Cybersecurity – Results: No Common Noise



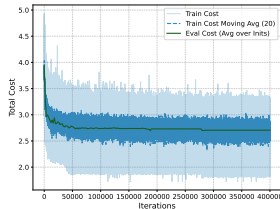
Evaluation trajectories for SyncFull starting from 4 different initializations.



SyncFull



SyncGreedy



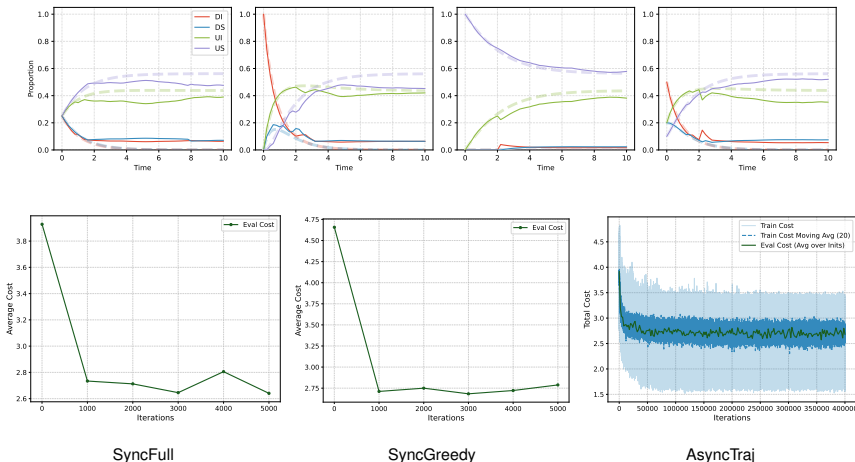
AsyncTraj

Ex. 1: Cybersecurity – Results: With Common Noise

The attack intensity v_H oscillates according to a stochastic process:

$$v_H^{n+1} = \text{proj}_{[0.7\bar{v}_H, 1.3\bar{v}_H]}(v_H^n + \epsilon_n^0), \quad \epsilon_n^0 \sim \mathcal{N}(0, \sigma^2)$$

with $\bar{v}_H = 0.6$ and $\sigma = 0.05$. Solvers remain robust to these shocks.



Value function:

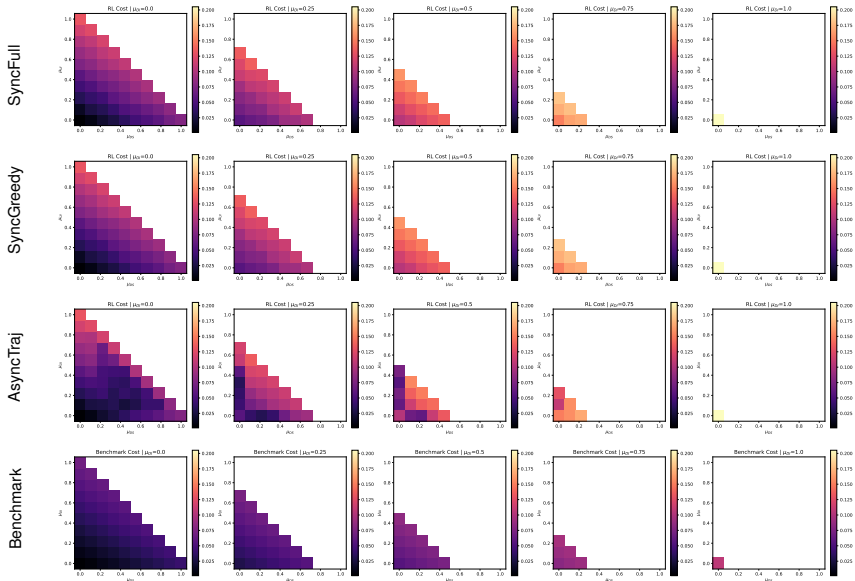
$$\bar{J}^*(\mu) = V^*(\mu) = \min_{\tilde{a}} Q^*(\mu, \tilde{a})$$

is a function of the 4D distribution $\mu \in \mathfrak{G}$.

Visualization:

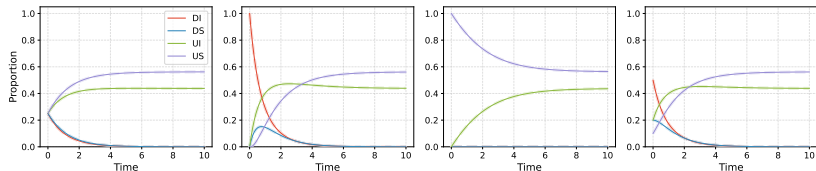
- ▶ Define 2D slices by fixing one state, e.g., $\mu_{DI} = \delta \in \{0, 0.25, 0.5, 0.75, 1\}$.
- ▶ Vary (μ_{DS}, μ_{UI}) in $[0, 1 - \delta]^2$ s.t. $\mu_{DS} + \mu_{UI} \leq 1 - \delta$.
- ▶ Remaining mass is $\mu_{US} = 1 - (\mu_{DI} + \mu_{DS} + \mu_{UI})$.
- ▶ This allows plotting the 4D simplex on a sequence of 2D triangles.

Ex. 1: Cybersecurity – Tabular Q vs. Benchmark (With CN)

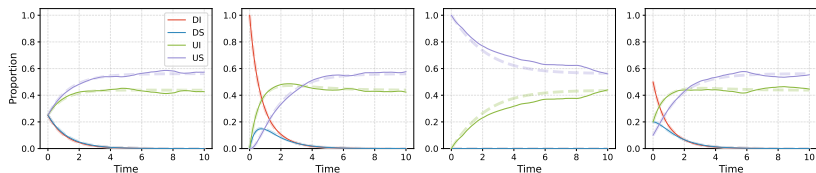


Ex. 1: Cybersecurity – DDPG

- ▶ **Continuous Space:** Action space $\mathcal{P}(A)^S \subset [0, 1]^4$ (defense intensity per state).
- ▶ **Architecture:** Actor/Critic with 2 layers of 32 neurons.
- ▶ **Result:** Trajectories are smoother and closer to benchmark.



DDPG (No CN)



DDPG (With CN)

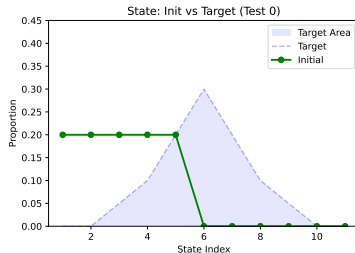
Ex. 2: Discrete Distribution Planning – Setup

Objective: Transport agents to match a target distribution with minimal movement.

- ▶ **States:** $S = \{1, \dots, 11\}$.
- ▶ **Actions:** $A = \{-1, 0, 1\}$ (Left, Stay, Right).
- ▶ **Dynamics:** $x_{n+1} = \text{proj}_S(x_n + a_n + \varepsilon_{n+1}^0)$.
- ▶ **Cost Function** (Social Cost):

$$\bar{f}(\mu_n, \tilde{a}_n) = \mathbb{E}_{x \sim \mu_n}[|a_n(x)|] + \lambda \|\mu_n - \mu_{\text{target}}\|_2^2$$

- ▶ **Target Profile** μ_{target} : A “Gaussian-like” bump centered at state 6.



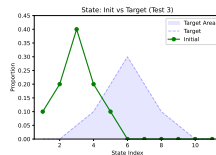
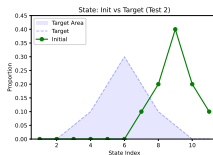
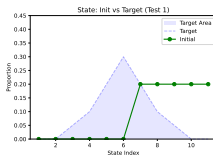
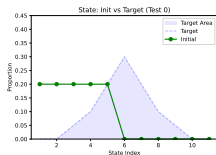
Ex. 2: Discrete Distribution Planning – Setup

Simulation Parameters:

Param	Val	Param	Val
N_{states}	11	T	50
CN Noise	± 1	Prob	0.05
γ	0.95	λ	5.0

DDPG Hyperparameters:

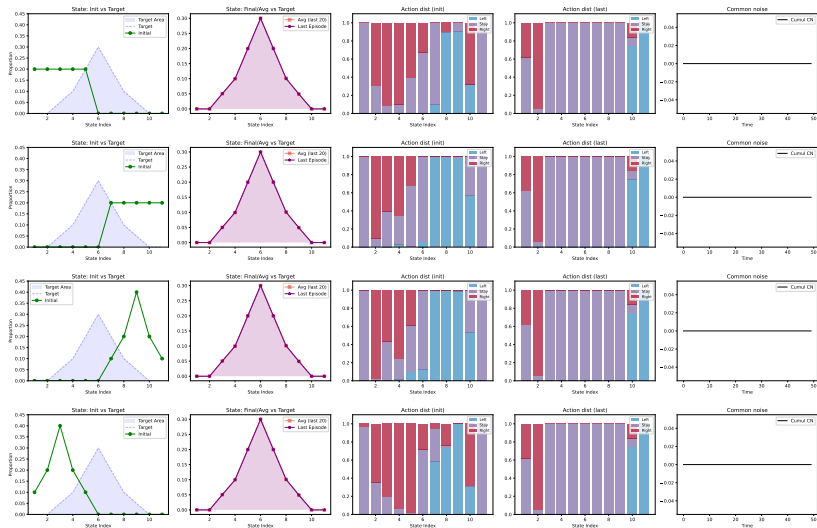
- **Architecture:** MLPs 2 layers, 128 neurons each.
- **Optimizer:** Adam (LR $\approx 10^{-3}$).
- **Target Update:** $\tau \approx 10^{-2}$.
- **Exploration:** Gaussian noise on $\tilde{a}$.



Initial distributions tested in Ex. 2.

Ex. 2: Discrete Distribution Planning – Results Without Common Noise

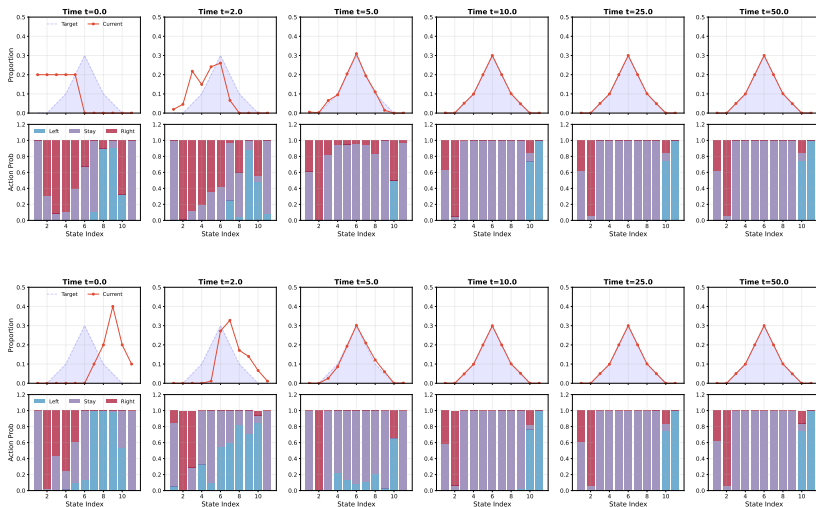
Evaluation from 4 different initial distributions (DDPG):



Columns: Initial State, Final State, Initial Action, Terminal Action, Common Noise (= 0).

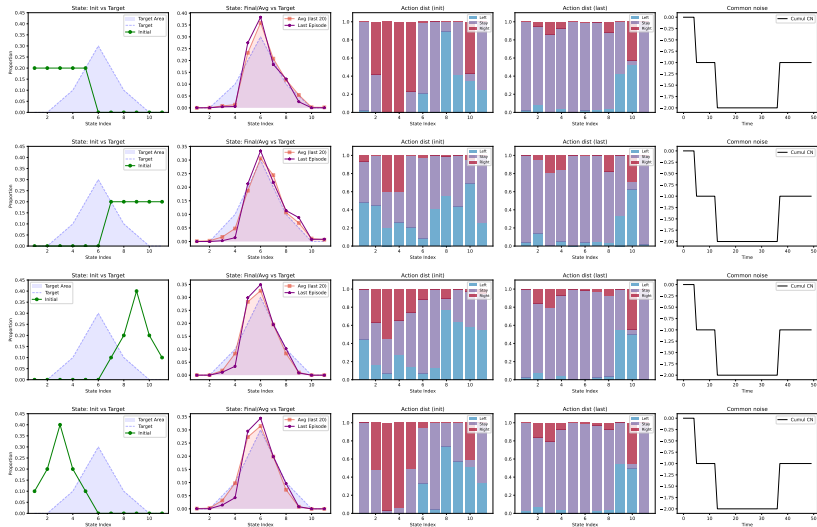
Ex. 2: Discrete Distribution Planning – Results Without Common Noise

Snapshots of the distribution evolution (no common noise):



Ex. 2: Discrete Distribution Planning – Results With Common Noise

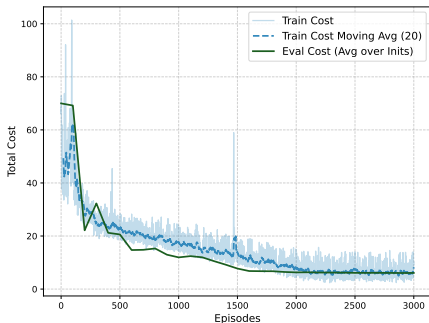
Robustness to stochastic shocks $\varepsilon^0 \in \{-1, 0, 1\}$ (DDPG):



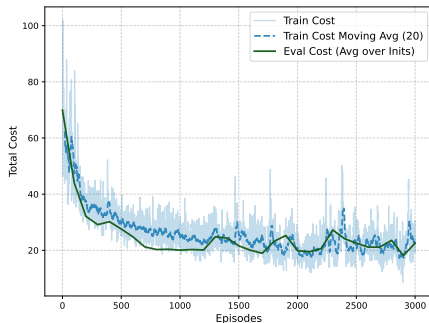
Columns: Initial State, Final State, Initial Action, Terminal Action, Common Noise.

Ex. 2: Discrete Distribution Planning – Learning Progress

DDPG training progress in both deterministic and stochastic environments:



No Common Noise



With Common Noise

- ▶ **Efficiency:** DDPG learns effective policies within ~ 2000 episodes.
- ▶ **Robustness:** Works even with common noise.

- ▶ **MFMDP viewpoint**: population-process MDPs [GG11, GGB12]; common-noise/open-loop MFC formulations [CLT23a, MP22]; dynamic programming and mean field RL [CLT23b, GGWX23, MP23].
- ▶ **Tabular Q-learning**: simplex discretization and convergence [CLT19a]; mean field Q-learning variants [GGWX20, CTSK21, AFL22a, AFLZ23].
- ▶ **Continuous-time and model-free MFC**: continuous-time Q-learning [WY25]; model-free policy gradient and actor-critic methods [MPR26, BK25, FGL⁺25, PW25].
- ▶ **Model-based, decentralized, robust, safe**: model-based MF-MARL [PKB23a, PKB23b, HYH24, HHK24]; decentralized network approaches [GGWX21a, GGWX24]; robust/social-control and safe mean-field MARL [XCW⁺25, JPJ⁺24].

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This part is mostly based on:

Carmona, Lauriere, Tan: [arXiv:1910.04295](#)

and lecture notes

Carmona & Lauriere: Mean Field Reinforcement Learning
([arXiv:2607.01525](#))

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Representative Agent Dynamics: For $n \geq 0$:

$$x_{n+1} = Ax_n + \bar{A}\bar{x}_n + Bu_n + \bar{B}\bar{u}_n + \varepsilon_{n+1}^0 + \varepsilon_{n+1}$$

where:

- ▶ $x_n \in \mathbb{R}^d$: state, $u_n \in \mathbb{R}^\ell$: action.
- ▶ $\bar{x}_n = \mathbb{E}[x_n | \mathcal{F}_n^0]$ and $\bar{u}_n = \mathbb{E}[u_n | \mathcal{F}_n^0]$: conditional means.
- ▶ ε_{n+1}^0 (common) and ε_{n+1} (idiosyncratic) noise terms.

Mean Field Cost: for $\mathbf{u} = (u_n)_{n \geq 0}$

$$J(\mathbf{u}) = \mathbb{E} \left[\sum_{n \geq 0} \gamma^n f(x_n, \bar{x}_n, u_n, \bar{u}_n) \right]$$

where f is of “tracking” form:

$$\begin{aligned} f(x, \bar{x}, u, \bar{u}) &= (x - \bar{x})^\top Q(x - \bar{x}) + \bar{x}^\top (Q + \bar{Q})\bar{x} \\ &\quad + (u - \bar{u})^\top R(u - \bar{u}) + \bar{u}^\top (R + \bar{R})\bar{u}. \end{aligned}$$

Theorem (Optimal Control)

The optimal control u_n^ is in feedback form and linear in $(x_n, \bar{x}_n)$:*

$$u_n^* = -K^*(x_n - \bar{x}_n) - L^*\bar{x}_n$$

where the optimal parameters are:

$$K^* = \frac{1}{2}R^{-1}B^\top P, \quad L^* = \frac{1}{2}(R + \bar{R})^{-1}(B + \bar{B})^\top \bar{P}$$

The matrices $P, \bar{P} \in \mathbb{R}^{d \times d}$ satisfy the discrete-time Riccati equations:

$$P = \gamma(A^\top P + 2Q)(A - BR^{-1}B^\top P/2)$$

$$\bar{P} = \gamma(\tilde{A}^\top \bar{P} + 2\tilde{Q})(\tilde{A} - \tilde{B}\tilde{R}^{-1}\tilde{B}^\top \bar{P}/2)$$

with $\tilde{A} = A + \bar{A}$, $\tilde{B} = B + \bar{B}$, $\tilde{Q} = Q + \bar{Q}$, $\tilde{R} = R + \bar{R}$.

Once the optimal gains (K^, L^*) are known, how does agent i apply the control?*

1. If the MFC mean $\bar{x}_n$ is known for all n (e.g., fixed initial condition and no common noise):

$$u_n^i = -K^*(x_n^i - \bar{x}_n) - L^*\bar{x}_n$$

The control is **decentralized**: agent i only needs to observe x_n^i .

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2. If $\bar{x}_n$ is unknown (e.g., due to common noise):

$$u_n^i = -K^*(x_n^i - \hat{x}_n^N) - L^*\hat{x}_n^N, \quad \text{with } \hat{x}_n^N = \frac{1}{N} \sum_{j=1}^N x_n^j$$

The control is **not fully decentralized** as it requires the empirical mean $\hat{x}_n^N$. However, in the LQMFC case, it only requires the mean, not the full distribution.

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The control is **not fully decentralized** as it requires the empirical mean $\hat{x}_n^N$. However, in the LQMFC case, it only requires the mean, not the full distribution.

Note: In general MFC (non-LQ), optimal control might depend on the **full distribution** in complex ways, not just the mean.

Cost Decomposition

Idea: Since the optimal control is linear, use Policy Gradient (PG) to search directly for $\theta = (K, L) \in \Theta$.

$$C(\theta) := J(\mathbf{u}^\theta).$$

Cost Decomposition

Idea: Since the optimal control is linear, use Policy Gradient (PG) to search directly for $\theta = (K, L) \in \Theta$.

$$C(\theta) := J(\mathbf{u}^\theta).$$

The cost $C(\theta)$ and decomposes as:

$$C(\theta) = C_y(K) + C_z(L)$$

where C_y (idiosyncratic) and C_z (aggregate) are standard LQR costs:

$$C_y(K) = \mathbb{E} \left[\sum_{n \geq 0} \gamma^n (y_n^\top (Q + K^\top R K) y_n) \right] = \text{Tr}((Q + K^\top R K) \Sigma_K)$$

$$C_z(L) = \mathbb{E} \left[\sum_{n \geq 0} \gamma^n (z_n^\top (\tilde{Q} + L^\top \tilde{R} L) z_n) \right] = \text{Tr}((\tilde{Q} + L^\top \tilde{R} L) \Sigma_L)$$

where the **state variance matrices** are defined as:

$$\Sigma_K = \mathbb{E} \left[\sum_{n \geq 0} \gamma^n y_n y_n^\top \right], \quad \Sigma_L = \mathbb{E} \left[\sum_{n \geq 0} \gamma^n z_n z_n^\top \right]$$

Theorem (Policy Gradient Expression)

For $\theta = (K, L) \in \Theta$, the gradient of the cost $C(\theta)$ is:

$$\nabla_K C(\theta) = 2E_K \Sigma_K, \quad \nabla_L C(\theta) = 2E_L \Sigma_L$$

where $E_K = (R + \gamma B^\top P_K^y B)K - \gamma B^\top P_K^y A$, and Σ_K is the state variance matrix.

Remarks:

- ▶ Gradient depends on the state variance matrix Σ_K and E_K , which depend on the model parameters. This is not model-free.
- ▶ Mirror results for standard LQR (Fazel et al., 2018).
- ▶ Fundamental for both analytical convergence and model-free implementations.
- ▶ Set of admissible policies: $\Theta := \{\theta = (K, L) \in \mathbb{R}^{\ell \times d} \times \mathbb{R}^{\ell \times d} \mid \gamma \|A - BK\|^2 < 1, \gamma \|A + \bar{A} - (B + \bar{B})L\|^2 < 1\}$

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Setup: Full knowledge of model parameters $(A, \bar{A}, B, \bar{B}, Q, \bar{Q}, R, \bar{R})$.

Algorithm: Exact Policy Gradient for MFC

Input: Initial policy $\theta^{(0)} = (K^{(0)}, L^{(0)}) \in \Theta$, learning rate η .

Output: Optimal policy θ^* .

```
1 for  $k = 0, 1, 2, \dots$  do  
2   Compute  $\nabla_K C(K^{(k)})$  and  $\nabla_L C(L^{(k)})$   
3   Update  $K^{(k+1)} \leftarrow K^{(k)} - \eta \nabla_K C(K^{(k)})$   
4   Update  $L^{(k+1)} \leftarrow L^{(k)} - \eta \nabla_L C(L^{(k)})$ 
```

The convergence analysis relies on two key properties of the cost functions:

1. Polyak-Lojasiewicz (PL) Condition:

$$\|\nabla C(\theta)\|^2 \geq \nu_{pl}(C(\theta) - C(\theta^*))$$

Even though the cost is non-convex, the gradient norm bounds the optimality gap.

2. Local Smoothness: The cost is smooth in the direction of the gradient descent for small enough steps:

$$C(\theta^{(k+1)}) - C(\theta^{(k)}) \leq \langle \nabla C(\theta^{(k)}), \theta^{(k+1)} - \theta^{(k)} \rangle + \frac{L_{loc}}{2} \|\theta^{(k+1)} - \theta^{(k)}\|^2$$

Theorem (Convergence of Exact PG)

*Under standard assumptions, if the learning rate $\eta > 0$ is sufficiently small, the costs converge **linearly** to the optimum:*

$$C(\theta^{(k)}) - C(\theta^*) \leq (1 - \eta\nu_{pl})^k (C(\theta^{(0)}) - C(\theta^*))$$

where $\nu_{pl} > 0$ is the PL constant.

Corollary (Iteration Complexity)

To reach a precision $\varepsilon > 0$, i.e., $C(\theta^{(k)}) - C(\theta^) \leq \varepsilon$, it is sufficient to take:*

$$k \geq \frac{1}{\eta\nu_{pl}} \log \left(\frac{C(\theta^{(0)}) - C(\theta^*)}{\varepsilon} \right)$$

1. **Admissibility:** Prove that for η small enough, the updated parameter $\theta^{(k+1)}$ remains in the set of stabilizing policies Θ .
2. **Descent Guarantee:** Use **local smoothness** to bound the decrease in cost at each step:

$$C(\theta^{(k+1)}) - C(\theta^{(k)}) \leq -\eta(1 - \text{bias})\|\nabla C(\theta^{(k)})\|^2$$

3. **Optimality Gap:** Invoke the **PL condition** to link the gradient norm to the distance from the global minimum:

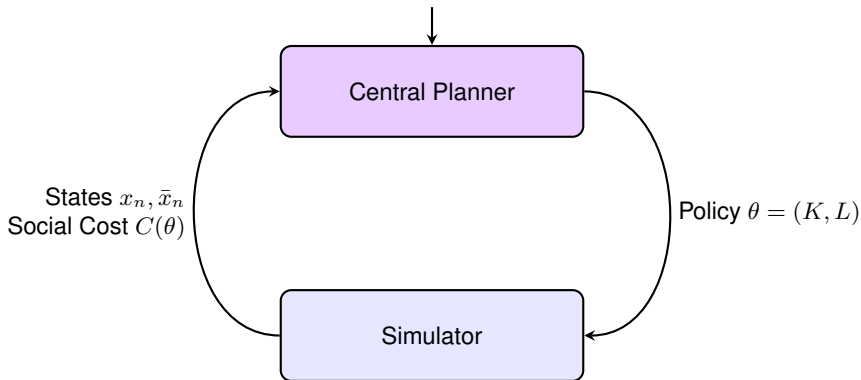
$$C(\theta^{(k+1)}) - C(\theta^{(k)}) \leq -\eta\nu_{pl}(C(\theta^{(k)}) - C(\theta^*))$$

4. **Iteration:** Re-arrange terms to show the geometric decay of the optimality gap.

What about model-free PG?

The LQMFC Reinforcement Learning Paradigm

Goal: Minimize $C(\theta)$ without knowing $(A, \bar{A}, B, \bar{B})$



Challenge: We only have access to **cost evaluations** based on **trajectories**.

Simulator: MKV-based or (finite) Population-based.

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MKV Simulator $\mathcal{S}_{MKV}^T$:

- ▶ Fix policy $\theta = (K, L)$
- ▶ Fix number of time steps T
- ▶ Simulate a trajectory following the McKean-Vlasov (= mean field) dynamics
- ▶ Return the truncated cost $\tilde{C}^T(\theta) = \sum_{n=0}^{T-1} \gamma^n c_n(\theta)$

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Zero-th order optimization:

- ▶ Estimate the gradient based only on samples of the cost;
- ▶ Plug this estimate in the PG algorithm instead of the true gradient.
- ▶ Uses $2M$ samples uniformly distributed on the unit sphere of radius τ .

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Algorithm: MKV-Based Gradient Estimation

Input: Parameter $\theta = (K, L)$, horizon T , M perturbations, radius τ .

Output: $\tilde{\nabla}^{T,M,\tau}(\theta)$.

- 1 **for** $i = 1, \dots, M$ **do**
- 2 Sample $v_i^{(idy)}, v_i^{(com)} \sim \text{Uniform}(\mathbb{S}_\tau)$
- 3 Set $\theta_i = (K + v_i^{(idy)}, L + v_i^{(com)})$
- 4 Receive $\tilde{C}^T(\theta_i)$ from $\mathcal{S}_{MKV}^T(\theta_i)$
- 5 **Return** $\tilde{\nabla}_K^{T,M,\tau}(\theta) = \frac{\ell d}{\tau^2} \frac{1}{M} \sum_{i=1}^M \tilde{C}^T(\theta_i) v_i^{(idy)}, \tilde{\nabla}_L^{T,M,\tau}(\theta) = \frac{\ell d}{\tau^2} \frac{1}{M} \sum_{i=1}^M \tilde{C}^T(\theta_i) v_i^{(com)}.$

Intuition for Zeroth-Order Optimization

Idea: Estimate the gradient of a function f by taking a random unit vector v and scaling it by $f(x + \tau v)$. The expectation is proportional to the gradient of a *smoothed* version of f .

Lemma 2.1 of Flaxman et al. (2005)

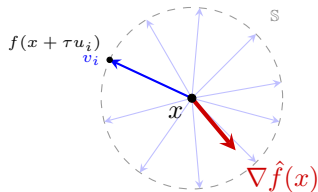
The expectation over the sphere is proportional to the smoothed gradient:

$$\mathbb{E}_{v \in \mathbb{S}}[f(x + \tau v)v] = \frac{\tau}{d} \nabla \hat{f}(x)$$

where $\hat{f}$ is the smoothed function defined as:

$$\hat{f}(x) = \mathbb{E}_{u \in \mathbb{B}}[f(x + \tau u)]$$

where $\mathbb{B}$ is the unit ball in $\mathbb{R}^d$ and u is uniform over the ball.



Intuition: Sampling on the sphere $\mathbb{S}_\tau$ gives the direction of the gradient of the smoothed function $\hat{f}$.

Theorem (Convergence with MKV Simulator)

For a target precision $\varepsilon > 0$, if η is small enough and (T, M, τ) satisfy:

- ▶ $\tau \lesssim \varepsilon$ (Small perturbation radius)
- ▶ $T \gtrsim \log(1/\varepsilon)$ (Sufficiently long horizon)
- ▶ $M \gtrsim \text{poly}(d, \ell, \delta^{-1})$ (High-probability concentration)

Then with probability $\geq 1 - \delta$, after $k \geq \frac{2}{\eta \nu_{pl}} \log(\frac{C_0}{\varepsilon})$ steps, we have:

$$C(\theta^{(k)}) - C(\theta^*) \leq \varepsilon$$

Note: We recover **linear convergence** up to the precision level ε despite using only noisy cost evaluations from the MKV simulator.

Sketch of Proof: Error Decomposition

The proof decomposes the error $\|\tilde{\nabla} - \nabla C\|$ into three terms:

1. **Smoothing Error:** Bias introduced by estimating the gradient of a smoothed version of the cost
 - ▶ Controlled by τ .
2. **Truncation Error:** Bias from approximating the infinite-horizon cost $C(\theta)$ with a finite-horizon sum $\tilde{C}^T$
 - ▶ Controlled by $\gamma^T \rightarrow 0$ as $T \rightarrow +\infty$.
3. **Sampling Error:** Variance from using M samples and stochastic noise in the MKV simulation
 - ▶ Controlled by M via concentration inequalities.

Combine these with the **PL condition** and **local smoothness** to show that the noisy update still follows the direction of descent.

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Algorithm: Population-Based PG

Population Simulator $\mathcal{S}_{pop}^{T,N}$: Simulates N agents following the same policy θ .

$$\tilde{C}^{T,N}(\theta) = \sum_{n=0}^{T-1} \gamma^n \left(\frac{1}{N} \sum_{j=1}^N f(X_n^{(j),\theta}, \bar{X}_n^{N,\theta}, u_n^{(j),\theta}, \bar{u}_n^{N,\theta}) \right)$$

Sources of error:

- ▶ trajectory truncation: T
- ▶ zero-order gradient estimation: τ, M
- ▶ mean field approximation by finite population: N

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- ▶ zero-order gradient estimation: τ, M
- ▶ mean field approximation by finite population: N

Algorithm: Population-Based Gradient Estimation

Input: Parameter $\theta = (K, L)$, horizon T , population size N , M perturbations, radius τ .

Output: $\tilde{\nabla}^{T,N,M,\tau}(\theta)$.

- 1 **for** $i = 1, \dots, M$ **do**
 - 2 Sample $v_i^{(idy)}, v_i^{(com)} \sim \text{Uniform}(\mathbb{S}_\tau)$
 - 3 Set $\theta_i = (K + v_i^{(idy)}, L + v_i^{(com)})$
 - 4 Receive $\tilde{C}^{T,N}(\theta_i)$ from $\mathcal{S}_{pop}^{T,N}(\theta_i)$
 - 5 Return gradients using zeroth-order formula as in MKV case.
-

Theorem (Convergence with Population Simulator)

If the population size N is sufficiently large, specifically:

$$N \geq \mathcal{O} \left(\frac{C_0}{\varepsilon \eta \nu_{pl}} \right)$$

and hyperparameters (τ, T, M) are tuned as in the MKV case, then with high probability:

$$C(\theta^{(k)}) - C(\theta^*) \leq \varepsilon$$

Consequence: A finite population can learn a policy that is approx. optimal for the **infinite-population** MF limit.

Key step: Approximation of the MF behavior by N agents.

Variance Identities: The noise and initial variances for the N -agent system are shifted by $1/N$:

$$\begin{aligned}\Sigma^{1,N} &= (1 - 1/N)\Sigma^1, & \Sigma^{0,N} &= \Sigma^0 + \Sigma^1/N \\ \Sigma_{y_0}^N &= (1 - 1/N)\Sigma_{y_0}, & \Sigma_{z_0}^N &= \Sigma_{z_0} + \Sigma_{y_0}/N\end{aligned}$$

Cost Difference: The difference between the social cost C^N and the MF cost C scales as $O(1/N)$. More precisely (see paper / lecture notes for notations):

$$|C^N(\theta) - C(\theta)| = \left| \frac{1}{N} \left\langle P_L^z - P_K^y, \Sigma_{y_0} + \frac{\gamma}{1-\gamma} \Sigma^1 \right\rangle_{tr} \right|$$

which leads to the bound:

$$|C^N(\theta) - C(\theta)| \leq \frac{1}{N} (\|P_K^y\| + \|P_L^z\|) \frac{\max\{Tr(\Sigma_{y_0}), Tr(\Sigma^1)\}}{1-\gamma}$$

Hence: The bias introduced by the finite population disappears as $N \rightarrow \infty$.

General Case: Conditional Propagation of Chaos

See Carmona and Delarue (2018), Volume II, for more details.

Theorem 2.12 *Within the above framework and under assumption **Conditional MKV SDE**, the system of particles (2.3) with $(X_0^1, \dots, X_0^N)$ as initial condition has a unique solution for every $N \geq 1$. It is denoted by $((X_t^{N,i})_{0 \leq t \leq T})_{i=1, \dots, N}$. Moreover,*

$$\lim_{N \rightarrow \infty} \left(\max_{1 \leq i \leq N} \mathbb{E} \left[\sup_{0 \leq t \leq T} |X_t^{N,i} - \underline{X}_t^i|^2 \right] + \sup_{0 \leq t \leq T} \mathbb{E} \left[W_2 \left(\frac{1}{N} \sum_{i=1}^N \delta_{X_t^{N,i}}, \mathcal{L}^1(\underline{X}_t^1) \right)^2 \right] \right) = 0.$$

When $\mathbb{E}^1[|X_0^1|^q] < \infty$ for some $q > 4$, there exists a constant C , only depending on T , $\mathbb{E}^1[|X_0^1|^q]$ and the Lipschitz constants of b , σ and σ^0 , such that:

$$\max_{1 \leq i \leq N} \mathbb{E} \left[\sup_{0 \leq t \leq T} |X_t^{N,i} - \underline{X}_t^i|^2 \right] + \sup_{0 \leq t \leq T} \mathbb{E} \left[W_2 \left(\frac{1}{N} \sum_{i=1}^N \delta_{X_t^{N,i}}, \mathcal{L}^1(\underline{X}_t^1) \right)^2 \right] \leq C \epsilon_N,$$

where $(\epsilon_N)_{N \geq 1}$ satisfies:

$$\epsilon_N = \begin{cases} N^{-1/2}, & \text{if } d < 4, \\ N^{-1/2} \log N, & \text{if } d = 4, \\ N^{-2/d}, & \text{if } d > 4. \end{cases} \quad (2.7)$$

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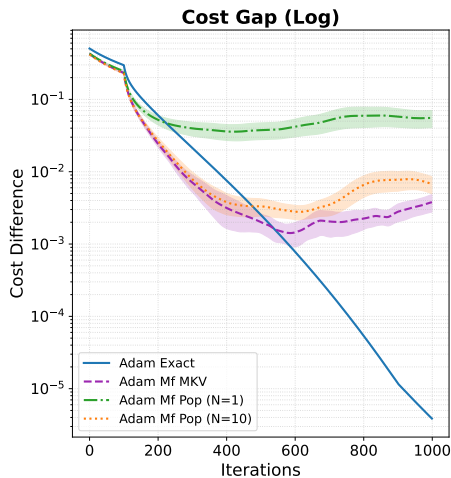
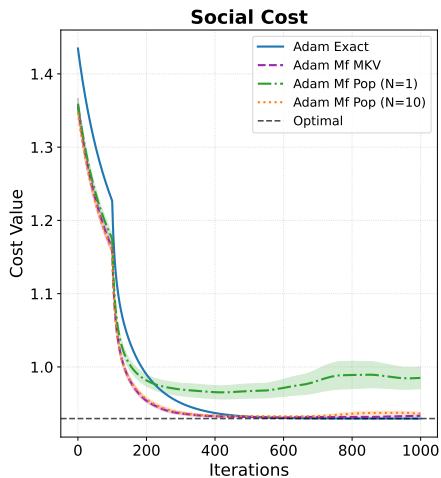
Model: Scalar LQ-MFC with idiosyncratic and common noise.

- ▶ **Learner:** Central planner using Adam for parameter updates.
- ▶ **Comparison:**
 - ▶ Exact PG
 - ▶ MF-MKV
 - ▶ MF-Pop with $N = 1$ and 10

Parameter	Value	Hyperparameter	Value
Dynamics $(a, \bar{a})$	$(0.5, 0.5)$	Learning rate η	0.01
Control $(b, \bar{b})$	$(0.5, 0.5)$	Radius τ	0.1
State cost $(q, \bar{q})$	$(0.5, 0.5)$	Horizon T	10
Control cost $(r, \bar{r})$	$(0.5, 0.5)$	Directions M	100
Discount factor γ	0.9	Iterations	1000
Variances σ^2	0.01	Seeds	10

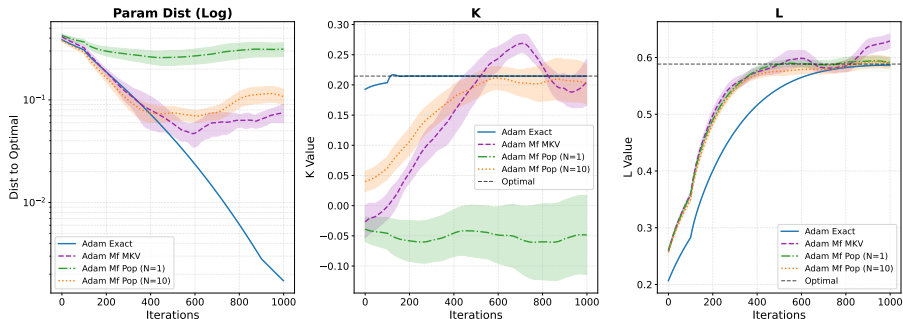
Learning rate: For practical efficiency and stability, we adapt the learning rate $\eta = \eta^{(k)}$ during iterations, using Adam optimizer.

Numerical Results: Cost Convergence



- ▶ Convergence to the theoretical optimal cost (black dashed line).
- ▶ **MKV simulator** (Purple dashed) closely follows the **Exact gradient** (Blue).
- ▶ **Pop. simulator**: High var. and bad convergence for $N = 1$; improve for $N = 10$.

Numerical Results: Parameter Convergence



- ▶ Convergence of parameters (K, L) toward (K^*, L^*) .
- ▶ Left: Distance to optimum. Middle/Right: K/L .
- ▶ Stable convergence across all model-free estimators.
- ▶ But for $N = 1$, K is not learned... and cannot be learned.

- ▶ **LQ mean field control:** policy-gradient analyses [[CLT19b](#), [CLT19c](#), [WHYW21a](#)].
- ▶ **Actor-critic and model-free variants:** LQ MFG actor-critic and PG [[FYCW19a](#), [FYCW19b](#), [uZZMB20a](#), [WHYW21b](#)]; model-free social-control learning [[WZZ20](#), [LLX24](#)].
- ▶ **Social optimization and robustness:** LQG social optimum by RL [[XWS25](#)]; robust mean field social control [[XCW⁺25](#)]; zero-sum mean-field type games [[CHLT20a](#), [CHLT21a](#), [CHLT20b](#), [CHLT21b](#)].
- ▶ **Recent PG directions:** model-free PG beyond LQ structure [[MPR26](#), [FGL⁺25](#)]; actor-critic with moment neural networks [[PW25](#)]; independent RL for cooperative-competitive agents [[uZKLB24a](#)].

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This part is mostly based on:

“Learning in Mean Field Games: A Survey”,
Lauriere, Perrin, Perolat, Girgin, Muller, Elie, Geist, Pietquin
(arXiv:2205.12944)

Two key operators define the Mean Field Game dynamics:

- ▶ **Mean Field Operator (MF)**: Maps a policy π to the sequence of distributions $\mu = \text{MF}(\pi)$ it generates.
- ▶ **Best Response Operator (BR)**: Maps a distribution flow μ to the set of policies $\pi \in \text{BR}(\mu)$ that minimize the individual cost $J(\pi; \mu)$.

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MFG Nash Equilibrium (MFNE)

A pair $(\hat{\pi}, \hat{\mu})$ is an MFNE if and only if $\hat{\pi} \in \text{BR}(\hat{\mu})$ and $\hat{\mu} = \text{MF}(\hat{\pi})$.

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This can be rewritten as a fixed point over policies or distributions:

$$\hat{\pi} \in \text{BR}(\text{MF}(\hat{\pi}))$$

or (if the BR is unique)

$$\hat{\mu} = \text{MF}(\text{BR}(\hat{\mu}))$$

Based on the fixed point formulation, we can define iterative algorithms.

- ▶ **Fixed Point Iteration:** Alternates directly between BR and MF updates:

$$\begin{aligned}\pi_{k+1} &\in \text{BR}(\mu_k) \\ \mu_{k+1} &= \text{MF}(\pi_{k+1})\end{aligned}$$

- ▶ *Warning:* Often fails to converge due to **oscillations**. BR operator $\text{BR}(\mu)$ might not be unique.

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- ▶ *Warning:* Often fails to converge due to **oscillations**. BR operator $\text{BR}(\mu)$ might not be unique.
- ▶ **Policy Iteration for MFG:** Intertwines standard MDP Policy Evaluation and Improvement with MF updates:

$$\begin{aligned}Q_{k+1} &= Q^{\mu_k, \pi_k} && \text{(Policy Evaluation)} \\ \pi_{k+1} &\in \arg \min Q_{k+1} && \text{(Policy Improvement)} \\ \mu_{k+1} &= \text{MF}(\pi_{k+1}) && \text{(Mean Field Update)}\end{aligned}$$

To prevent oscillations and ensure uniqueness of the Best Response, we use two main techniques:

1. **Entropic Regularization**: Modify the cost by adding an entropy penalty $\eta \log(\pi(a|x))$. This makes the optimal policy a softmin policy (smooth), ensuring uniqueness. Tuning the temperature gives smoothness.
2. **Averaging Iterations**: Do not jump directly to the new BR/MF. Average the history of past strategies or distributions.
 - ▶ **Fictitious Play (FP)**: Averages policies or distributions over iterations.
 - ▶ **Online Mirror Descent (OMD)**: Maintains a cumulative Q-function to softly update the policy.

Fictitious Play (FP)

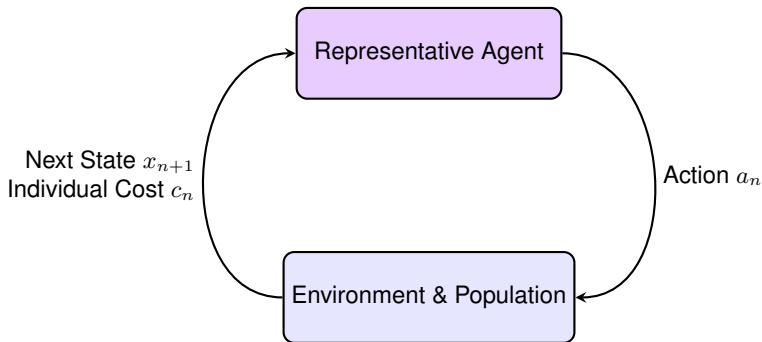
Averages past distributions.

1. $\pi_{k+1} \in \text{BR}(\bar{\mu}_k)$
2. $\mu_{k+1} = \text{MF}(\pi_{k+1})$
3. $\bar{\mu}_{k+1} = \frac{k}{k+1}\bar{\mu}_k + \frac{1}{k+1}\mu_{k+1}$

Online Mirror Descent (OMD)

Accumulates evaluating Q-functions.

1. $Q_{k+1} = Q^{\mu_k, \pi_k}$
2. $\bar{Q}_{k+1} = \bar{Q}_k + \alpha Q_{k+1}$
3. $\pi_{k+1} \propto \exp(-\bar{Q}_{k+1}/\tau)$
4. $\mu_{k+1} = \text{MF}(\pi_{k+1})$



The population mean field μ evolves internally. The agent **only** observes its individual state and cost.

How do we implement the fixed point iterations without knowing the model?

- ▶ **Inner Loop (RL):** Since μ is fixed (or $\bar{\mu}$ in FP), the problem is a standard MDP.
 - ▶ We can compute the **Best Response** using standard model-free RL algorithms (Q-learning, DQN, PPO, DDPG) running in a simulated environment with frozen population behavior.
 - ▶ For **Policy Iteration** / **OMD**, we replace the exact evaluator Q^{μ_k, π_k} with a model-free **Policy Evaluation** step (e.g., TD-learning).

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- ▶ **Outer Loop (MF Update):**
 - ▶ Once the inner loop RL converges, we deploy the new policy π_{k+1} (MKV simulator or Population simulator).
 - ▶ We obtain the new mean field μ_{k+1} .
 - ▶ Apply the FP averaging or OMD accumulation, and repeat the outer loop.

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- ▶ **Mixture:** the two loops can be “mixed” by adjusting the number of iterations in each loop.

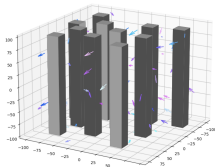
MFG Numerical Example: Flocking

Setting [PLP⁺21]:

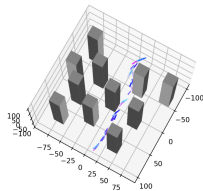
- ▶ Population of birds adjusting their acceleration to match the average velocity of their neighbors.
- ▶ Continuous state space: positions and velocities. Continuous action space: acceleration.
- ▶ **Model-Free Approach:** Since state/action spaces are continuous, we cannot use tabular methods.

Algorithm: Deep Fictitious Play

- ▶ **Distribution μ :** Modeled with a Normalizing Flow.
- ▶ **Policy π :** Learned via Deep RL (Soft Actor-Critic – SAC).



Initial distribution



At convergence

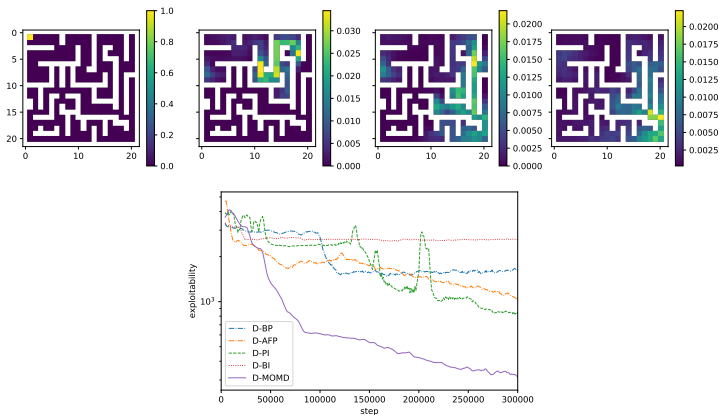
MFG Numerical Example: Crowd Exiting a Maze

Setting [LPG⁺22]:

- ▶ A large crowd of agents tries to exit a maze while avoiding congested areas.
- ▶ Demonstrates the scalability of advanced algorithms for Mean Field Games.

Algorithm: Deep Online Mirror Descent (OMD) and Deep Fictitious Play (FP)

- ▶ Uses Neural Networks to approximate the cumulative Q-function and the population distribution.



More details:

- ▶ **Papers:**

- ▶ “Learning in Mean Field Games: A Survey”, Lauriere et al. (arXiv:2205.12944)
- ▶ References at the end

- ▶ **Personal webpage:**

- ▶ slides
- ▶ codes

- ▶ **Surveys, software, benchmarks:** learning MFG survey [[LPP⁺](#)]; scalable deep RL algorithms and OpenSpiel examples [[LPG⁺22](#), [LLL⁺19](#)]; Bench-MFG [[MSG⁺26](#)].
- ▶ **Fixed-point, value, and Q-learning:** value iteration and Q-learning [[GHXZ19](#), [AKS20](#), [AKS19](#), [AKS23](#), [AKS21](#), [GHXZ23](#)].
- ▶ **Fictitious play, OMD, policy iteration:** learning and averaging schemes [[PPL⁺20](#), [PPE⁺21a](#), [CCG21](#), [CT21](#), [LST21](#)].
- ▶ **Deep RL, MARL, inverse RL:** deep RL and distribution embeddings [[EPL⁺20](#), [CK21a](#), [PLP⁺21](#), [MSW⁺25](#)]; mean-field MARL and inverse RL [[YLL⁺18](#), [YYT⁺17](#), [CLK21](#), [CZLH22](#), [RKP⁺24](#), [ZKBB23](#)].

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- ▶ **Mean Field RL Paradigm:** Central planner viewpoint for cooperative agents using the Mean Field Markov Decision Process (MFMDP) framework.
- ▶ **Discrete Settings:**
 - ▶ **Model-Free Q-learning:** Tabular approach with convergence guarantees.
 - ▶ **Exploration/Exploitation:** Sweep-based vs trajectory-based updates.
- ▶ **Continuous and Deep RL:**
 - ▶ **MF-DDPG:** Scaling to continuous state-action spaces using deep NN.
 - ▶ **Scalability:** Handling complex environments.
- ▶ **Linear-Quadratic MFC:**
 - ▶ **Global Convergence:** Proofs for Policy Gradient (PG) methods.
 - ▶ Decoupling of idiosyncratic and aggregate states via PL condition.
- ▶ **Practical Implementation:**
 - ▶ Model-free PG using zeroth-order estimation (MKV and Pop. simulators).
 - ▶ Robustness of Adam-based updates in stochastic environments.

► **Problem Settings:**

- Finite horizon, ergodic/average cost, ...
- Entropy-regularized, risk-sensitive objectives ...

► **Beyond Homogeneity and Symmetry:**

- Survey: [An overview of some extensions of mean field games beyond perfect homogeneity and anonymity](#), L., Science China Information Sciences (2026)
- Multi-population models: several types of agents.
- Graphon control: non-exchangeable agents on large graphs.
- ...

► **Information Structure:**

- Class of policies: centralized vs decentralized.
- Partial observation, imperfect information, ...

► **Beyond Pure Cooperation:**

- MFG: Non-cooperative settings (Nash equilibria).
- MF Type Games / MF Teams: Nash eq. between cooperative teams.
- MF Control Games: Limit of infinitely many competing teams.
- Mixed Individual/Population Games
- ...

▶ **Theoretical RL:**

- ▶ Theoretical convergence for Deep RL for MFC?
- ▶ Rates of convergence?

▶ **Mean Field Representation:**

- ▶ Efficient approximation of probability measures?
- ▶ How does it interact with the RL algorithm?

▶ **Extensions of MFC:**

- ▶ RL for the variants mentioned in the previous slide?

▶ **Real-World Scenarios:**

- ▶ Concrete applications?
- ▶ Data-driven approaches?
- ▶ Ongoing progress:
 - ▶ Inverse Problems for Costs and Controls in LQG MFGs via Mean Field Trajectories, Lambrecht & Lauriere, CDC'26
 - ▶ Neural Parameter Calibration for Finite-State Mean Field Games, Thöni et al. (2026)

- ▶ **Multiple populations and heterogeneity:** multi-population/team learning [SPTH20, SSM18, GSPTH20, PPE⁺22, YVCD20, uZMB23]; graphon learning [CK21b, FCK22, CKK22, FCK23, HWYZ23, ZTWY24].
- ▶ **Beyond pure cooperation:** mean field type/control games and mixed games [ADF⁺22, ADF⁺23, CDDL23]; zero-sum settings [CHLT20b, CHLT21b, GAT23, uZKLB24b].
- ▶ **Correlated and equilibrium learning:** policy-space response oracles and correlated equilibria [MRE⁺22, MER⁺22, WCL⁺22].
- ▶ **Partial observation:** partially observed MFC/MFG and decentralized policies [CHFK23, CHFK24b, STCP20, CHFK24a, YLZW23].

- ▶ **Model-based and sample-efficient RL:** model-based MF-MARL [PKB23b, HYH24, HHK24]; decentralized learning [YAY22, STCP22, YCGH23, YGH24].
- ▶ **Inverse and imitation learning:** inverse RL for MFG/MFC [YYT⁺18, CLK21, CZLH22, RKP⁺24, AKS24]; traffic-flow inverse RL [MCD⁺24].
- ▶ **Representation and diagnostics:** distribution approximation in flocking and continuous MFGs [PLP⁺21, MSW⁺25]; OpenSpiel and Bench-MFG environments [LLL⁺19, MSG⁺26].
- ▶ **Regularization, exploration, and safety:** exploration noise [DV21]; regularized/stable learning [ASA⁺23, MQW⁺22]; robust and safe social control [XCW⁺25, JPJ⁺24].

Thank you!

Slides:



<https://mlauriere.github.io/slides-public/2026-changsha-mfml-tutorial.pdf>

Feel free to reach out if you have questions: mathieu.lauriere@nyu.edu

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